NIHR | National Institute for Health and Care Research

Summary

Reference Number	NIHR305248
Lead Applicant	Mrs Sarah Firman
Research Title	Impairment of muscle mass, muscle function and quality of life in phenylketonuria: optimising protein intakes safely, effectively and feasibly
Plain English Summary	Phenylketonuria (PKU) is an inherited metabolic condition that leads to the buildup of an amino acid (a building block of protein) called phenylalanine. All babies born in the UK are tested for PKU at birth. High phenylalanine levels can cause severe brain damage; however, early treatment prevents this. Treatment is for life, and involves reducing the amount of protein in the diet (to usually less than 15 grams daily) to limit phenylalanine intake. The treatment also involves taking medical supplements (called PKU supplements) that contain all amino acids, except for phenylalanine. For adults with PKU, we do not know the ideal amount of protein needed each day (from foods and PKU supplements). We know from studies in healthy adults that dietary protein plays an important role in keeping muscles healthy. It is important to get the amount of protein right for adults with PKU. Too little protein could lead to muscle problems later in life, whereas too much protein, through taking more PKU supplements, could make the diet too hard to follow. We have planned two studies: **Study 1: Do people with PKU suffer from poor muscle health, and is it related to diet?** We will measure how a PKU diet affects muscle mass (the amount of muscle someone has) and muscle function (how well the muscle works) in adults with PKU. Participants will be asked to keep a record of their food intake for four days, and to wear an accelerometer on their thigh, a match box size device that monitors movement, for a week before visiting a centre for the following tests: 1. Exercises to measure muscle function: jumping, climbing steps and grip strength 2. Muscle mass measurements: a DXA scan (full-body x-ray scan), bioelectrical impedance analysis (similar to scans used in some gyms, four small stickers are placed on the hands and feet), and skin thickness in different places on the body using a hand-held device (skinfold caliper) 3. Blood tests to measure levels of different proteins 4. A bloodspot test to measure phenylalanine (similar to a finger prick test for blood sugar) 5. Questionnaires about diet, quality of life, and exercise **Study 2: Does increasing protein intake improve muscle mass and function in PKU?** Participants in Study 1 who are found to have a low muscle function will be invited to join a second study. This study will see if taking more protein over 12 weeks improves muscle mass and/or function. Half the participants will take their normal amount of protein, and the other half will be asked to take more protein (through increased PKU supplements). Everyone will be advised to do regular exercises. During the 12 weeks, participants will keep a diary to record their diet and exercise routine. They will be asked to send in a bloodspot test every 14 days, similar to their current clinical care. After 12 weeks, the participants will repeat the same tests as Study 1. They will also have an ultrasound scan to measure the size of their arm and thigh muscles at the beginning and end of Study 2. Finally, participants and Dietitians will be asked to give feedback on the studies to help improve further research. People with PKU have been involved in the study design and will continue to help with patient information leaflets, questionnaires, and sharing the results with the PKU community. We hope that these studies will help design a larger study looking at how diet can be

	adjusted to improve muscle mass and function in adults with PKU. The results will also guide Doctors and Dietitians to ensure adults with PKU maintain good health as they age.

Our purpose for collecting this data is to communicate with you about your application and have the necessary information to evaluate you for a grant. The data we collect here is collected in the public interest. Information provided here may be subject to Freedom of Information requests.

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The NIHR privacy policy includes further information including ways we may use your data, our contact details and details on your individual rights regarding how your data is used. Your data may be shared across the NIHR, including with other coordinating centres, to allow the application to be managed and for statistical analysis, and with external grant reviewers as part of the process for managing the allocation of a grant. This notice is under constant review and will be updated and / or revised based on that review as appropriate.

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1. Application Summary Information

Host organisation (which will administer any award)
King's College London

Partner organisation	Guy's and St Thomas' NHS Foundation Trust

Research Title	Impairment of muscle mass, muscle function and quality of life in phenylketonuria: optimising protein intakes safely, effectively and feasibly

Partnership Fellowship Please select the relevant Partnership Fellowship.
N/A

Proposed WTE Please select which WTE option you would prefer
1.00

Duration (months) The duration for your award is based on your chosen WTE
36

Proposed start date if grant awarded	01/09/2025

Research Type	Feasibility study, Other (please specify)

Please provide more details on your selection above.
Observational study

2. Applicant CV

A CV for the Applicant is a mandatory component of this application; please ensure this is completed fully.
Please note that some of the CV questions are automatically completed from information you have provided in the 'Update CV' section of the 'Manage My Details' page. These can be updated via the ARAMIS Home Page or by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Degrees and Professional Qualifications

From	To	Qualification	Subject	Country	Institution	Class
10/2019	07/2020	Postgraduate Certificate	Clinical Research	United Kingdom (England)	King's College London	Distinction
01/2011	12/2012	Master of Dietetics	Dietetics	New Zealand	University of Otago	Distinction
02/2008	11/2010	BSc	Human Nutrition	New Zealand	University of Otago	First Class Honours

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
12/2021	-	Senior Specialist Adult Inherited Metabolic Diseases Dietitian (1.0 WTE)	Guy's and St Thomas' NHS Foundation Trust	10
05/2020	-	Dietitian representative on the Medical and Scientific Advisory Board (Voluntary)	Global DARE Foundation	5
10/2019	11/2021	Senior Specialist Adult Inherited Metabolic Diseases Dietitian (0.5 WTE)	Guy's and St Thomas' NHS Foundation Trust	5
10/2019	11/2021	HEE/NIHR Pre-Doctoral Clinical Academic fellow (0.5 WTE)	Guy's and St Thomas' NHS Foundation Trust	50
01/2019	10/2022	Dietetic Representative for the Trust Clinical Guidelines Group (Voluntary)	Guy's and St Thomas' NHS Foundation Trust	0
05/2017	10/2019	Senior Specialist Adult Inherited Metabolic Diseases Dietitian (1.0 WTE)	Guy's and St Thomas' NHS Foundation Trust	0
01/2017	05/2017	Locum Adult Inherited Metabolic Diseases Dietitian	Guy's and St Thomas' NHS Foundation Trust	0
07/2016	12/2022	Locum Dietitian (Renal & Critical Care)	Guy's and St Thomas' NHS Foundation Trust	0
05/2016	09/2016	Locum Renal Dietitian (Part-time)	Epsom and St Helier University Hospitals NHS Trust	0
04/2015	03/2016	Professional Development Convener for the Auckland, New Zealand, Dietitians Branch	Voluntary for the Auckland Dietitians Branch	0
04/2013	04/2016	Renal Dietitian	Waitemata District Health Board, New Zealand	0
02/2013	04/2013	Dietitian Volunteer	Waitemata District Health Board, New Zealand	0

Present and previous positions
(including from/to dates, job title and employer)

From	To	Position	Organisation	WTE % Research
04/2011	08/2012	Research Assistant (Part-time)	University of Otago	10

Research Grants held

Title of Grant	Role in the Research Grant	Total Grant £	Source of Grant (Organisation Name)	Duration (In Months)	Start Date of Funding	End Date of Funding
HEE / NIHR Pre-doctoral Clinical Academic Fellowship 2019	PGCert in Clinical Research, Cochrane online modules, and courses in statistics and data analysis. PPIE training and established a PPIE group. Conducted two literature reviews: lead on protocol design, data collection and publication (DOI:10.1016/j.clnu.2020.11.003, DOI:10.1136/bmjopen-2021-049883, DOI:10.1016/j.clnu.2022.02.010) Conducted national survey: lead on project design, recruitment, data collection, analysis and publication (DOI:10.1111/jhn.13015)	66951	HEE/NIHR	24	01/10/2019	30/09/2021
Individual Staff Development Award	Travel grants to present research abstracts at the British Inherited Metabolic Diseases Group conference (Nottingham, 2018) and the Society for the Study of Inborn Errors of Metabolism international conference (Rotterdam, 2019)	1442	Guy's and St Thomas' Charity	1	21/06/2018	06/09/2019

Publication Record
This should display the authors, date, article title and journal name for each publication added. This information can be edited via the “My Research Outputs” page of your ARAMIS account.

Publications
Daly, A., Adam, S., Allen, H., Ash, J., Dale, C., Dixon, M., Dunlop, C., Ellerton, C., Evans, S., **Firman, S.**, Ford, S., Freedman, F., Gribben, J., Howe, S., Khan, F., McDonald, J., McStravick, N., Nguyen, P., Oxley, N., Skeath, R., Simpson, E., Terry, A., Woodall, A., White, L. and MacDonald, A.. (2022) ‘UK Dietary Practices for Tyrosinaemias: Time for Change.’, *Nutrients*, 14(24).
Gama, MI., Adam, S., Adams, S., Allen, H., Ashmore, C., Bailey, S., Cochrane, B., Dale, C., Daly, A., De Sousa, G., Donald, S., Dunlop, C., Ellerton, C., Evans, S., **Firman, S.**, Ford, S., Freedman, F., French, M., Gaff, L., Gribben, J., Grimsley, A., Herlihy, I., Hill, M., Khan, F., McStravick, N., Millington, C., Moran, N., Newby, C., Nguyen, P., Purves, J., Pinto, A., Rocha, JC., Skeath, R., Skelton, A., Tapley, S., Woodall, A., Young, C. and MacDonald, A.. (2022) ‘Suitability and Allocation of Protein-Containing Foods According to Protein Tolerance in PKU: A 2022 UK National Consensus.’, *Nutrients*, 14(23).
Firman, SJ., Ramachandran, R. and Whelan, K.. (2022) ‘Knowledge, perceptions and behaviours regarding dietary management of adults living with phenylketonuria. ’, *Journal of human nutrition and dietetics : the official journal of the British Dietetic Association*.
Firman, SJ., Ramachandran, R., Whelan, K., Witard, OC. and O'Keeffe, M.. (2022) ‘Protein status in phenylketonuria: A scoping review.’, *Clinical nutrition (Edinburgh, Scotland)*, 41(4).

Publication Record

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Publications
Wood, G., Pinto, A., Evans, S., Daly, A., Adams, S., Costelloe, S., Gribben, J., Ellerton, C., Emm, A., **Firman, S.**, Ford, S., French, M., Gaff, L., Giuliano, E., Hill, M., Hunjan, I., Newby, C., Mackenzie, A., Pereira, R., Prescott, C., Robertson, L., Seabert, H., Skeath, R., Tapley, S., Terry, A., Tooke, A., van Wyk, K., White, FJ., White, L., Woodall, A., Rocha, JC. and MacDonald, A.. (2021) ‘Special Low Protein Foods Prescribed in England for PKU Patients: An Analysis of Prescribing Patterns and Cost. ’, *Nutrients*, 13(11).
Firman, S., Ramachandran, R., Whelan, K., Witard, OC. and O'Keeffe, M.. (2021) ‘Protein status of people with phenylketonuria: a scoping review protocol.’, *BMJ open*, 11(9).
Firman, S., Witard, OC., O'Keeffe, M. and Ramachandran, R.. (2020) ‘Dietary protein and protein substitute requirements in adults with phenylketonuria: A review of the clinical guidelines.’, *Clinical nutrition (Edinburgh, Scotland)*, 40(3).
Evans, S., Adam, S., Adams, S., Allen, H., Ashmore, C., Bailey, S., Banks, J., Churchill, H., Cochrane, B., Cook, J., Dale, C., Daly, A., Dixon, M., Dunlop, C., Ellerton, C., Emm, A., **Firman, S.**, Ford, S., French, M., Gribben, J., Grimsley, A., Herlihy, I., Hill, M., Judd, S., Lang, K., Males, J., McDonald, J., McStravick, N., Millington, C., Newby, C., Noble, C., Pereira, R., Pinto, A., Robertson, L., Robotham, A., Ross, K., Singleton, K., Skeath, R., Terry, A., Van Wyk, K., White, F., White, L., Wildgoose, J., Woodall, A. and MacDonald, A.. (2020) ‘Uniformity of Food Protein Interpretation Amongst Dietitians for Patients with Phenylketonuria (PKU): 2020 UK National Consensus Statements. ’, *Nutrients*, 12(8).
Green, B., Browne, R., **Firman, S.**, Hill, M., Rahman, Y., Kaalund Hansen, K., Adam, S., Skeath, R., Hallam, P., Herlihy, I., Jenkinson, F., Nicol, C., Adams, S., Gaff, L., Donald, S., Dawson, C., Robertson, L., Fitzachary, C., Chan, H., Slabbert, A., Dunlop, C., Cozens, A., Newby, C., Bittle, V., Hubbard, G. and Stratton, R.. (2019) ‘Nutritional and Metabolic Characteristics of UK Adult Phenylketonuria Patients with Varying Dietary Adherence.’, *Nutrients*, 11(10).
Green, B., Rahman, Y., **Firman, S.**, Adam, S., Jenkinson, F., Nicol, C., Adams, S., Dawson, C., Robertson, L., Dunlop, C., Cozens, A., Hubbard, G. and Stratton, R.. (2019) ‘Improved Eating Behaviour and Nutrient Intake in Noncompliant Patients with Phenylketonuria after Reintroducing a Protein Substitute: Observations from a Multicentre Study.’, *Nutrients*, 11(9).

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
01/07/2020	King's College London	Distinction for Postgraduate Certificate in Clinical Research	Postgraduate student
01/09/2019	Guy's and St Thomas' Charity	Awarded a travel bursary to attend and present a research abstract at the Society for the Study of Inborn Errors of Metabolism Conference 2019 in Rotterdam, The Netherlands.	Research abstract presenter
01/06/2018	Guy's and St Thomas' Charity	Awarded a travel bursary to attend and present a research abstract at the British Inherited Metabolic Diseases Conference 2018 in Nottingham, UK.	Research abstract presenter
01/06/2015	Rotary Young-Person Leadership Award (RYLA)	Awarded a Rotary Young-Person Leadership Award: To attend a week-long	Leadership programme attendee

Relevant Prizes, Awards and other Academic Distinctions

Please note the information for additional prizes or awards is automatically populated from information provided in the "Update CV" section of the "Manage My Details" page, which you can update by following this link https://aramis.nihr.ac.uk/MyAccount/UpdateCV.aspx

Date (mm/yyyy)	Source	Prize	Role of Applicant
		leadership development programme designed to develop teamwork and communication skills, and build my potential as a leader.	
01/12/2012	Nestlé Healthcare	Nestlé Healthcare Nutrition Top Student Award: Awarded by Dietitians New Zealand to the top Dietetic student in New Zealand.	Dietetic student
01/12/2012	Abbott Nutrition Bernice Kelly Award	Awarded by Dietitians New Zealand to recognise excellence in an entry-level dietitian.	New graduate Dietitian
01/12/2012	University of Otago	Distinction for Master of Dietetics thesis	Master's student
01/12/2010	University of Otago	Awarded University of Otago Scholarship in Science	Undergraduate student
01/11/2007	University of Otago	Scholarship for Dux	Undergraduate student

ORCID iD 0000-0001-7257-700X

3. Applicant Research Background

Professional Background	Allied Health Professional

Profession (Allied Health Professional)	Dietitian

Do you have current / active professional registration? This registration must be with one of the **ICA approved regulatory bodies**
Yes

Professional registration number/pin Please supply a valid professional registration number/pin followed by the regulatory body (eg 12A3456B - NMC or PA12345 - HCPC)
DT28709

Please describe your research career to date

As the lead Senior Specialist Dietitian in one of only five nationally commissioned adult inherited metabolic diseases (IMDs) services in England, I manage over 1000 patients with complex nutritional needs, including phenylketonuria (PKU). Due to advances in neonatal screening and paediatric management, the number of patients transitioning to adult services has increased exponentially. Despite diet being the mainstay of treatment, there is limited evidence informing the dietary management of adults with PKU, and other IMDs, creating an urgent need for research that was rarely previously needed, a novel gap that I aim to fill. I aspire to become a clinical academic and research leader, optimising the health outcomes of adults living with IMDs.

Research experience

Pre-doctoral Clinical Academic Fellowship (PCAF)

I was awarded a HEE/NIHR PCAF in 2019 that afforded me the opportunity to extend my research knowledge and skills. I completed a PGCert in Clinical Research, including quantitative and qualitative research design, data analysis and critical appraisal, and during this PCAF, published 4 first-author peer-reviewed full publications, totaling 10 full publications in my NHS career (see 'Publication Record'). I expanded my research skills by undertaking the following initiatives:

Evidence review and synthesis

I completed the "Cochrane: Conducting an Intervention Review" online course, and attended several courses at King's College London (KCL) on systematic searches and writing literature reviews. I consolidated these skills when I led on a narrative review and scoping review on protein requirements and protein status in PKU, both published in "Clinical Nutrition" the highest impact journal in Nutrition & Dietetics[1,2]. My reviews are cited in multiple publications including two consensus papers in PKU[3,4].

Research design and analysis

I completed an online course in Logistic Regression and attended courses on research development and analysis (Good Clinical Practice, academic writing and SPSS data analysis).

I employed these skills when conducting a national survey to investigate knowledge, perceptions and dietary

behaviours of adults with PKU, published in the Journal of Human Nutrition & Dietetics[5]. I led on the conceptualisation, design, ethics approval, questionnaire development in consultation with service users, analysis, and publication.

Clinical research involvement

The Adult IMD service at Guy's and St Thomas' Hospital (GSTT) has a longstanding commitment to research, collaborating with industry and other IMD services. I was the lead Dietitian for the following studies:

- Multi-centre trial of a new PKU supplement (Nutricia, 2017-2018).
  Role: Recruitment, randomisation, data collection, data interpretation, and contribution to publication (see 'Publication Record').
- Multi-centre retrospective study led by Professor Anita MacDonald, Birmingham Children's Hospital, investigating metabolic control of patients with PKU across Europe and the UK (2019-2022).
  Role: Recruitment, data interpretation and I will co-author prospective publications.

Supporting research within my department and discipline

I am passionate about Dietitians as leaders in clinical and research activities. I led on the design and delivery of several audits and service evaluations. Over the last five years, the IMD Dietetic service has presented 14 abstracts at prestigious Metabolic conferences, of which I was first author or supervised a team member to lead on the abstract. Undertaking my PhD will enable me to further support dietitians and other clinical academic researchers.

Conference presentations

I have experience in presenting research abstracts at national and international conferences, including the International Congress of Inborn Errors of Metabolism 2021 (Sydney, Australia), Society for the Study of Inborn Errors of Metabolism (SSIEM) 2019 (Rotterdam, Netherlands), National Society of PKU conference 2019 (UK), and British Inherited Metabolic Diseases conference 2018 and 2022 (UK), and have three abstracts submitted to present at SSIEM 2024 (Porto). I was a panelist at the UK and Ireland Metabolic Dietitians meeting in 2023 contributing to discussions on protein requirements in IMDs, and have been invited to open this event in 2024, presenting on my work in protein requirements and protein status in PKU.

Patient and public involvement and engagement (PPIE)

I established the first PKU PPIE group at GSTT in 2020 and have facilitated multiple meetings. This group has been pivotal in the development of my doctoral research proposal (see 'Patient and Public Involvement'). I have undertaken training in PPIE, and aim to do further training in my PhD fellowship.

MSc Research Thesis

I completed my Masters (Distinction) in 2012 where my thesis assessed the feasibility of computer-tailored nutrition communications in primary care. I was responsible for recruitment, conducting semi-structured interviews with patients and healthcare professionals, and quantitative and qualitative data analysis.

<u>Professional experience</u>

Clinical skills and leadership

I am passionate about our NHS and developing our services, having worked in the NHS for eight years. As the lead Senior Metabolic Dietitian at GSTT, I provide supervision and clinical leadership to the junior Dietitians. I started in 2017 as the only Dietitian in the IMD service, and over the last seven years I have developed the service

to a team of four Dietitians and one Dietetic Assistant. I have led on multiple service development projects including creating standardised care pathways, and advancing the role of Dietitians by establishing the first Dietitian-led IMD clinics under GSTT, and developing training competencies and clinical resources to support Dietitians managing complex IMD.

I am a visiting lecturer at KCL, teaching on the Advanced Diet Therapy training module for dietetic students.

National and international advisor

I have been involved in the development and publication of national consensus guidelines for PKU and other IMD conditions, either as the lead contributor from GSTT or supervising a member of staff to support their involvement and contribution to the publications[3,6,7,8]. I am a co-author of the IMD chapter in the Manual of Dietetic Practice (7th Edition, in press).

GSTT is a centre of excellence for a rare condition, Adult Refsum Disease (ARD). I am the international lead Dietitian for ARD, developing the international dietary guidelines and providing clinical expertise to Dietitians worldwide, and sit on the Advisory Board for the Global DARE Foundation.

Membership

As a member of the British Metabolic Dietitians Group, I have been actively involved in advancing clinical practice in IMD through contributing to the development of consensus statements and care pathways.

Have you already registered for a PhD?	No

Has this application, or a similar application, previously been submitted to this or any other funding body?
Yes

If Yes, provide details

Title: Impairment of muscle mass, muscle function and quality of life in phenylketonuria: prevalence and potential for dietary and lifestyle intervention

Funding body and scheme: NIHR DCAF

Outcome: Unsuccessful

This application, submitted in 2022, has been revised based on the feedback received. I scored highly for all the 'Applicant' sections (both reviewers providing an overall score of 5, excellent). Reviewers commented on the strength of the Patient and Public Involvement and Engagement (PPIE), but with a suggestion to increase PPIE members in the Project Advisory Group from one to 2-3 members. The supervisory team and host organisation were very highly regarded by the reviewers; however, there was recommendation to consider a further practice supervisor of the same professional background. The following feedback was provided for the research plan: greater consideration for the design of the intervention in relation to the MRC framework; incorporating further qualitative research to inform intervention development and delivery from both the participants' and healthcare professionals' (HCP) perspectives, and determining how exercise intervention fidelity would be assessed. With recommendations to include more qualitative research, further training on qualitative research was indicated.

This application differs from previous submissions in that the following amendments have been made in response to the above feedback:

1. **Project Advisory Group:** Three PPIE members will now be in the Advisory Group. A diverse PPIE will be selected to ensure members represent the views of the wider PKU community.

2. **Practice supervisors**: Dr Danielle Bear has joined my supervisory team as a member from the same

professional background as myself. Dr Bear is a former NIHR ICA awardee and since completing her NIHR PhD has been appointed as a Consultant Dietitian (GSTT) and Clinical Academic Lecturer in Nutritional Sciences (KCL) (see 'Supervision' section).

3. **Research plan**:
    a. **MRC/NIHR and TFA frameworks** - The MRC and NIHR framework informed the research proposal including incorporating the six core elements in the development and assessment of the intervention. Additionally, the Theoretical Framework of Acceptability (TFA) will now inform assessments of intervention acceptability. The initial study included establishing patients' acceptability of the intervention; however, the perspective of the HCP was not included. The revised research plan includes a focus group with Dietitians to include their perspectives.
    b. **Fidelity of physiotherapy intervention** - Collaborations with the physiotherapist has strengthened exercise intervention fidelity, with the rehabilitation treatment specification system informing the development of the intervention (see 'Research plan'). Each participant will receive the same written guidance and a one-to-one video induction session to ensure correct exercise technique and intensity. Having one designated physiotherapist delivering the exercise intervention will ensure consistency in its delivery. They will remain blinded to the dietary group allocation of participants. Throughout the 12-week intervention period, exercise type, frequency and intensity will be assessed to monitor adherence to the protocol.

Training and development: With incorporating further qualitative research in the feasibility study, I have updated my training plan to include courses on qualitative research design, delivery, and analysis offered by the Social Research Association (see 'Training and Development' section).

Contextual Factors

Please use this question to detail any contextual factors you wish to make the Selection Committee aware of. NIHR wants to know about any circumstances so that they may take them into consideration during the assessment of your application. Contextual factors may include:

- Career breaks due to parental leave, or periods of illness.
- Reduced time spent undertaking research due to a disability or caring responsibilities. This could include any physical or mental difficulty that may have impacted your research career. These are situations that have a significant impact on your ability to undertake research.
- Reduced opportunities to career support e.g. mentorship, and limited opportunities to undertake prior research and training.
- Impact of the COVID-19 pandemic on your research career.

Please also use this section to detail any other factors that may have impacted your research career not listed in the examples provided. The impact on your career to date will be specific to your particular circumstances but could include such impacts as limited opportunities to obtain grant funding or fewer publications. In general terms, mitigating factors should be significant, and relevant.

NIHR acknowledges that you may be reluctant, or uncomfortable disclosing relevant information that is sensitive. However, you should bear in mind that we are unable to take into account factors that you do not disclose. Please be assured that the information provided by you is sensitive and will be treated confidentially and in line with General Data and Protection Regulations (GDPR).

I was on maternity leave from November 2022 and returned to work in August 2023.

4. Plain English Summary of Research

Phenylketonuria (PKU) is an inherited metabolic condition that leads to the buildup of an amino acid (a building block of protein) called phenylalanine. All babies born in the UK are tested for PKU at birth. High phenylalanine levels can cause severe brain damage; however, early treatment prevents this. Treatment is for life, and involves reducing the amount of protein in the diet (to usually less than 15 grams daily) to limit phenylalanine intake. The treatment also involves taking medical supplements (called PKU supplements) that contain all amino acids, except for phenylalanine.

For adults with PKU, we do not know the ideal amount of protein needed each day (from foods and PKU supplements). We know from studies in healthy adults that dietary protein plays an important role in keeping muscles healthy. It is important to get the amount of protein right for adults with PKU. Too little protein could lead to muscle problems later in life, whereas too much protein, through taking more PKU supplements, could make the diet too hard to follow.

We have planned two studies:

Study 1: Do people with PKU suffer from poor muscle health, and is it related to diet?

We will measure how a PKU diet affects muscle mass (the amount of muscle someone has) and muscle function (how well the muscle works) in adults with PKU. Participants will be asked to keep a record of their food intake for four days, and to wear an accelerometer on their thigh, a match box size device that monitors movement, for a week before visiting a centre for the following tests:

1. Exercises to measure muscle function: jumping, climbing steps and grip strength
2. Muscle mass measurements: a DXA scan (full-body x-ray scan), bioelectrical impedance analysis (similar to scans used in some gyms, four small stickers are placed on the hands and feet), and skin thickness in different places on the body using a hand-held device (skinfold caliper)
3. Blood tests to measure levels of different proteins
4. A bloodspot test to measure phenylalanine (similar to a finger prick test for blood sugar)
5. Questionnaires about diet, quality of life, and exercise

Study 2: Does increasing protein intake improve muscle mass and function in PKU?

Participants in Study 1 who are found to have a low muscle function will be invited to join a second study. This study will see if taking more protein over 12 weeks improves muscle mass and/or function. Half the participants will take their normal amount of protein, and the other half will be asked to take more protein (through increased PKU supplements). Everyone will be advised to do regular exercises.

During the 12 weeks, participants will keep a diary to record their diet and exercise routine. They will be asked to send in a bloodspot test every 14 days, similar to their current clinical care.

After 12 weeks, the participants will repeat the same tests as Study 1. They will also have an ultrasound scan to measure the size of their arm and thigh muscles at the beginning and end of Study 2.

Finally, participants and Dietitians will be asked to give feedback on the studies to help improve further research.

People with PKU have been involved in the study design and will continue to help with patient information leaflets, questionnaires, and sharing the results with the PKU community.

We hope that these studies will help design a larger study looking at how diet can be adjusted to improve muscle mass and function in adults with PKU. The results will also guide Doctors and Dietitians to ensure adults with PKU

maintain good health as they age.

5. Scientific Abstract

Background

Phenylketonuria (PKU) is an inherited metabolic disorder of protein metabolism. Treatment involves limiting dietary protein to restrict phenylalanine intake and supplementation with protein substitutes providing all other amino acids. The optimal protein intake for adults with PKU is unknown. Current protein intake guidelines are extrapolated from healthy populations, using estimates based on nitrogen balance (NB) methodology, which have limitations in both healthy adults and in PKU. Protein requirements based on Indicator Amino Acid Oxidation (IAAO) are up to 55% higher for PKU. Establishing optimal protein intake in adults with PKU is essential as inadequate protein intake will likely compromise muscle mass and function, whereas excessive protein intake, through increased protein substitute doses, is a substantial burden to patients.

I will conduct two studies with the following aims and methods:

Study 1: Observational study

The aim will be to assess muscle mass and function of adults with PKU, and investigate whether impairments are associated with current dietary management practices.

A single-centre observational study will recruit 50 participants with PKU, aged ≥18 years following a stable management plan. Participants will complete a 4-day food diary and wear an accelerometer for 7 days prior to attending a test day. A range of gold standard and common clinical assessments of muscle mass and function with be conducted, including: handgrip strength, isokinetic dynamometry, 6-minute step test, vertical jump test, dual energy X-ray absorptiometry, bioelectrical impedance analysis, skinfold thickness, and various biochemical markers of protein status. All measurements will be compared to normative data. Dietary intake, physical activity and quality of life will also be assessed.

Participants on a PKU diet with a handgrip strength <25th centile of reference data will be invited to participate in Study 2.

Study 2: Feasibility study

The aim will be to examine the feasibility of a study investigating the impact of prescribing protein intake, based on NB-informed protein requirements (current practice) versus IAAO-informed protein requirements, on muscle mass and function.

A randomised control trial (RCT) feasibility study will recruit 24 participants. Eligible participants will be randomised into one of two arms for 12-weeks:

- **NB protein requirements (0.83g/kg/day)** plus home-based exercise programme
- **IAAO protein requirements (1.2g/kg/day)** plus home-based exercise programme

Assessment data from Study 1 will form baseline measurements for Study 2, and will be repeated at week 12. Additionally, muscle mass and thickness will be measured by upper and lower limb ultrasound at weeks 0 and 12. During the 12-weeks, participants will complete fortnightly bloodspots and physical activity questionnaires. Adherence to the interventions will be monitored through a weekly log. Feasibility measures, including recruitment, retention and intervention fidelity will be collected, and questionnaires and semi-structured interviews conducted to evaluate intervention acceptability.

Impact and dissemination

This research will contribute to the development of evidence-based dietary guidelines for adults with PKU. Findings will be published in peer-reviewed journals and presented at conferences. Study 2 will provide preliminary data to inform the design of a full-scale RCT investigating dietary and lifestyle factors modulating muscle mass and function in adults with PKU.

6. Detailed Research Plan

Background

Phenylketonuria (PKU) is a rare inherited metabolic disorder of protein metabolism with a prevalence of 1:10,000 in the UK. Left untreated, high serum phenylalanine causes significant irreversible neurocognitive disability. Dietary management remains the mainstay of treatment, consisting of lifelong adherence to a very low protein diet (typically <15 grams per day) and provision of up to 80% of protein intake from phenylalanine-free or low-phenylalanine protein substitutes[9].

1. Problems being addressed

Establishing optimal protein intake in adults with PKU: "Not too little, not too much"

With the onset of newborn screening for PKU in 1969, there is now an increasing adult and ageing PKU population requiring dietary guidance to support changing metabolic demands across the lifespan. In my narrative review, I highlighted that there is currently limited evidence informing protein requirement guidelines for adults with PKU[1].

Current protein intake guidance for adults with PKU is extrapolated from healthy population data, using estimates of protein requirements based on either nitrogen balance (NB) methodology or from younger adults with PKU[1]. The limitations associated with NB methodology are widely recognised[10]. Utilising Indicator Amino Acid Oxidation (IAAO) as a more contemporary approach to estimating protein requirements, protein requirements have been reported to be 39-55% higher in PKU, than recommendations informed by NB methodology[11]. Increased consumption of specialised protein substitutes would be required to achieve this higher protein intake; however, people on a PKU diet report poor palatability of these supplements, and adherence to a PKU diet has been shown to reduce with age[12-14].

The impact of the PKU diet on muscle mass and function, and by extension, risk of cardiometabolic disease and frailty in later life, is unknown[2,15]. Given that a PKU diet involves modifying sources of ingested protein, establishing the optimal protein intake in adults with PKU is essential because an inadequate protein intake will likely compromise muscle mass and function (encompassing strength, power and mobility), whereas excessive protein intake will result in a substantial prescribed supplement burden to patients.

2. Importance of research to improving health of service users and health care services in PKU

Adults with PKU are concerned about their muscle mass and function and impact on quality of life

Our PKU PPIE group identified that adults with PKU are concerned about the impact of the PKU diet on functional ability as they age. My nationwide questionnaire survey in 137 people with PKU identified that >50% have concerns for their long-term health, echoing the issues raised by our PPIE group[5]. Maintenance of muscle mass is crucial for supporting function, reducing cardiometabolic risk and frailty, and improving quality of life (QoL) with advancing age[16].

Following a very restrictive diet is a significant burden for people with PKU. In the UK, 73% of adults find PKU dietary management difficult and 39% do not take their protein substitutes as prescribed[12]. Poor dietary adherence has been reported to increase guilt and anxiety, thus negatively impacting health[13]. Research is therefore needed to establish the minimum dietary protein intake that translates to optimal muscle mass and function, without compromising phenylalanine control, to support healthy ageing in PKU.

Research is urgently needed to inform patient care

I surveyed Dietitians from nine UK-based metabolic centres and found varying protein prescribing practices. The UK Metabolic Dietitians Annual meeting 2023 included a discussion on 'Protein requirements – have we got it right?', for which I was a panelist. Concerns were raised regarding discrepancies in dietary management as a result of limited evidence. Current European PKU guidelines recommend the FAO/WHO/UNU safe minimum protein requirements informed by NB studies[9]; however, the recent IAAO informed protein requirements indicate this significantly underestimates requirements[11].

Current clinical practice is informed by guidelines without any scientific basis in the target patient group - to date no studies have investigated the impact of protein intake informed by NB *vs*. IAAO on the important endpoints of muscle mass and function in adults with PKU – and I want to change this.

With protein playing an essential role in skeletal muscle health[17], there is an urgent unmet need for research to better inform protein requirements for adults with PKU and how protein status is monitored in clinical practice.

3. Review of existing evidence

Timing

My narrative review highlighted a paucity of evidence regarding protein requirements for people with PKU, with significant omissions in adult specific recommendations[1]. There is an immediate need for research specifically for adults with PKU who, following successful neonatal screening programmes, are for the first time now entering middle-older adulthood, and are being advised lifelong PKU restricted diets. We must ensure dietary management leads to optimal health outcomes across the life course.

Relevance

Protein status in adults with PKU

I designed and published the protocol[18], and subsequently performed and published the first-ever scoping review to understand how dietary management translates to body protein status (encompassing anthropometric, biochemical and functional measures) in people with PKU[2]. I identified 70 studies that measured relevant markers; however, only one study assessed protein status as the primary outcome. The majority of studies were conducted in children and adolescents with PKU, with very few in adults. Taken together, this review highlights a significant gap in the current evidence and urgent need for research focused on understanding protein status in adults with PKU.

Relevant outcome measurements

In my scoping review, I identified a small number of studies with mixed findings regarding muscle mass in adults with PKU *vs*. healthy controls[2]. The only study to measure muscle function reported 38% of participants with PKU had grip strength <3rd percentile[2,19]. Furthermore, pre-albumin (biochemical marker of protein status) was lower in people with PKU[2]. Since publishing my scoping review, two recent studies have reported lower muscle mass and handgrip strength in adults with PKU compared to controls[20,21]; however, these studies were in a different healthcare system (Chile and Hungary) and did not address the questions raised in this research proposal. To date, no study has measured muscle mass, function and biochemical markers of protein status in adults with PKU, and investigated impacts on QoL. This multiple measurement approach is important for two reasons; first, because adequate muscle mass does not mean adequate muscle function[22] and it is muscle function that is important to people with PKU, and second, because Dietitians do not know the best measurement method to use in practice.

Relevant intervention

The optimal protein intake for adults with PKU remains unknown. No intervention studies have investigated the impact of recommending dietary protein intake based on NB-informed protein requirements (0.83g/kg/day) (standard current practice) *vs*. IAAO-informed protein requirements (1.2g/kg/day) on muscle mass and function in adults with PKU. I aim to establish protein requirements for adults with PKU by performing a feasibility study (as part of my PhD) and then undertaking a larger RCT (as part of a NIHR ACAF).

4. Research aims and objectives

This research proposal aims to:

(i) assess muscle mass, muscle function and biochemical parameters of protein status in adults with PKU;

(ii) investigate whether impairments in muscle mass, muscle function and biochemical parameters in adults with PKU are associated with PKU dietary management practices;

(iii) examine the feasibility of a study investigating the comparison of protein intake informed by NB (0.83g/kg/day) *vs.* IAAO (1.2g/kg/day) and its impact on muscle mass and function, and phenylalanine control in adults with PKU.

Objectives

I propose to undertake two studies:

Study 1: Observational study

- Primary objective: To investigate differences in muscle mass and function (anthropometric, functional and biochemical measurements) in adults with PKU *vs*. reference healthy population dataset.
- Secondary objective: To investigate associations between dietary factors (protein intake, protein substitute use), muscle mass and physical activity, and muscle function in adults with PKU.

Study 2: Feasibility study:

- To determine the impact of increasing prescribed protein intake and exercise training on muscle mass and function in adults with PKU who have compromised muscle strength.
- To establish the feasibility of implementing this complex intervention over 12 weeks in adults with PKU, with the view to designing a full-scale RCT.

5. Project Plan

Study 1: Observational study

Study design

A cross-sectional, observational study will be conducted over an 18-month period in adults with PKU enrolled from the Adult Metabolic Service at GSTT.

An observational study is critical to understanding the association between current dietary management practices on muscle mass and function in adults with PKU and to establish the most robust endpoints for use in future trials. The proposed research will be the first to simultaneously measure anthropometric, functional and biochemical parameters of protein status in an adult UK cohort with PKU.

Methods

Recruitment and consent

Fifty adults with PKU will be recruited. Following advice from our PPIE group, due to the large geographical area covered by the service, appointments to discuss the study will be virtual (video/telephone). Eligible patients will be sent a study information pack and participant consent will be returned via post. Additionally, to ensure equality of access to participation, travel and time taken away from work or other duties will be reimbursed.

Inclusion criteria:

- Diagnosed with PKU on newborn screening
- Followed a stable management plan for ≥12 months
- Aged ≥18 years
- Able to walk independently up at least 1 flight of stairs

Exclusion criteria:

- Plasma phenylalanine concentration <600 µmol/L without treatment (mild PKU)
- Pregnant, lactating or pre-conception (as dietary management and metabolic control differs)
- Other co-morbidities that could influence protein intake and digestion

Data collection

The protocol is outlined in Figure 1. Participants will attend a test day at the Centre for Human and Applied Physiological Sciences, King's College London (KCL) for ~3 hours. I will perform all assessments outlined below, and data will be entered into REDCap EDC.

Muscle function

Muscle function assessments will be the primary focus of this observational study as they were deemed most important by our PPIE group. I have selected the following assessments with normative reference data that are validated in physically active, adult populations[23–26]. With the exception of handgrip strength (measured in only two PKU studies)[19,21], none of these functional assessments have previously been measured in adults with PKU.

Handgrip strength – Widely recognised as a strong predictor of whole-body muscle strength[27] and prognostic of age-related health outcomes[28].

Isokinetic dynamometry – This technique measures dynamic lower limb muscular contraction and is the gold standard to assess muscle strength and power[29].

Whilst isokinetic dynamometry is gold standard, it cannot be easily used in the clinical setting. Measurements by isokinetic dynamometry will be compared to handgrip strength, a tool readily available in the clinical setting with potential for widespread use in PKU following this study.

6-minute step test (6MST) - Exercise capacity is indirectly correlated with functional ability in healthy adults, and is a key determinant of QoL and health outcomes[30]. The test will be performed on a single 20-cm step in accordance with the published protocol[25].

Vertical jump test on force platform - The vertical jump test provides a measurement of lower limb muscular power, an important physiological measure associated with the ability to perform activities of daily living and a sensitive measurement of muscular change in healthy adults[31]. Vertical ground reaction force will also be measured using a laboratory force plate, the gold standard for this measurement.

Anthropometry assessment

A range of gold standard and common clinical measures of anthropometry will be used. The correlation between gold standard (e.g. dual energy X-ray absorptiometry (DXA)) *vs.* common clinical measures (e.g. bioelectrical impedance analysis (BIA) and skinfolds) remain unknown in people with PKU, and can be measured here to inform assessment approaches in clinical practice.

DXA - the gold standard measurement of lean mass, will be used to assess whole body and regional lower limb lean mass. DXA measurements of lean mass closely correlate with measurements from MRI and CT, but at a lower cost and with only small doses of radiation[32].

BIA and skinfold thickness - these measurements provide estimates of fat-free mass and are common methods for assessing body composition in a clinical setting. BIA has been validated in children and adolescents with PKU[32], but not adults[2]. Assessing fat-free mass in adults with PKU using these easily accessible techniques is both practical and cost effective.

Height, weight, body mass index, and waist and thigh circumference will also be assessed.

Biochemical markers

Biochemical markers to assess protein status of people with PKU, including plasma amino acids, albumin, prealbumin, total protein, retinol-binding protein and related biomarkers, will be measured. On test days, venous blood samples (maximum of 20 ml) will be collected, processed, stored and analysed by the Clinical Biochemistry laboratory at GSTT.

Metabolic control – Management of PKU in adults aims for blood phenylalanine concentrations between 120-600 µmol/L. Dried bloodspots (Guthrie cards) will be collected on waking on Day -7 and the test day, in line with patients' usual methods for self-monitoring.

Dietary assessment

Food diary and Food Frequency Questionnaire (FFQ)

A four-day food diary will be recorded and analysed by Nutritics software to determine current dietary protein intake. In addition, the EPIC-FFQ, validated in adults within the UK[34], will be completed to gain insight into habitual intakes and compared to food diary data to identify whether FFQ is accurate to be routinely used in practice.

Physical activity assessment

Assessments of physical activity in adults with PKU are limited[2]. A range of physical activity measures will be recorded, including gold standard and clinical measures.

Accelerometery – Habitual activity will be recorded for 7 days prior to the test day using a thigh-worn activPAL™ accelerometer device, which our PPIE group regarded as feasible. This accelerometer quantifies free-living daily activities, accurately recording the duration and intensity of all activities[35].

International Physical Activity Questionnaire – Short Form (IPAQ-SF) - IPAQ-SF is a validated and widely used self-report questionnaire providing internationally comparable data on physical activity[36]. This questionnaire will be used alongside accelerometry to assess physical activity.

Health-related QoL

The PKU-QoL questionnaire has been validated in adults with PKU in the UK[37]. This questionnaire will be used to understand the impact of PKU on physical and mental health, psychosocial factors, and the challenges and barriers of adherence to a PKU-specific diet.

Other measurements – A questionnaire will be developed, and piloted with our PPIE group, to gather information on diagnosis, PKU management (dietary and pharmacological) and adherence, and demographic characteristics (e.g. age, sex, ethnicity) to ensure the study cohort is representative of the wider PKU population.

Data analysis

The following statistical analysis plan was developed in conjunction with KCL Statistical Support services. I will be responsible for all data analysis with ongoing and extensive support from the KCL Statistical Services and my supervisors.

Sample size

A sample size of 43 is required to detect differences in the primary measure (handgrip strength) in comparison to healthy reference population data. This sample size is sufficient to detect a difference in mean grip force z-score of -0.64 ± 1.26 (reported in Choukair et al, 2017) with 90% power and 5% significance from a two-sided one-sample t-test (with the sample taken from a limited population of 4000 individuals with PKU in the UK).

Up to 50 participants will be recruited to obtain complete data for at least 43 participants. This sample size will be sufficient to investigate the association between handgrip strength and other continuous variables such as total protein intake, muscle mass and physical activity. This sample size will achieve 83.5% power to detect moderate correlation between variables (Pearson's correlation coefficient >0.4) using a two-sided hypothesis test with 5% significance.

Analysis

Participant characteristics and muscle mass and function measures will be reported using n (%) for categorical data, and mean (±SD) or median (IQR) as appropriate for continuous data. Associations and differences between handgrip strength (continuous variable) and dietary factors, muscle mass and physical activity will be analysed using Pearson's correlation coefficient for continuous variables, unpaired t-test for binary variables and one-way

ANOVA for polychotomous variables with Bonferroni post hoc correction. Variable selection methods will be employed to select ~5 variables that explain the greatest variability in handgrip strength (or other outcomes of interest) prior to the use of linear regression models. All analysis will be conducted using SPSS Statistics and statistical significance will be $p<0.05$.

Study 2: Feasibility study

Study design

The feasibility study will be a RCT with two intervention arms conducted over 12-weeks (see Figure 2). A 12-week duration is frequently used in studies investigating the impact of protein intake and/or exercise training on muscle mass and functional outcomes[38].

The intervention is a complex intervention with the interacting components of diet and exercise, that are behaviour dependent. Therefore, the MRC/NIHR framework for complex interventions[39] guided the development of this research proposal, along with involvement from our PPIE group. The validated Theoretical Framework of Acceptability (TFA) will be used to inform assessments of intervention acceptability[40].

The feasibility of conducting a large RCT to assess the impact of protein intake on muscle mass and function in adults with PKU is unknown. A feasibility study will be important to ascertain the acceptability and practicality of the interventions, willingness to participate, retention rate, and to inform intervention development and delivery from both the patients' and healthcare professionals' perspectives.

It remains unclear which outcome(s) relating to muscle mass and function will be the most appropriate in a large clinical trial, and therefore, multiple gold standard (e.g. DXA and isokinetic dynamometry) and common clinical outcomes (e.g. BIA, ultrasound and handgrip strength) will be conducted in this feasibility study.

An exercise intervention will be included in both study arms, as exercise, especially resistance-exercise training (RET), has a fundamental role alongside dietary protein intake in improving muscle mass and function[38]. Furthermore, our PPIE group felt it was important to offer exercise to both groups, and in doing so, reducing confounding if habitual activity differed between groups.

Methods

Recruitment and consent

Eligible participants will be identified at the end of their observational study test day based on their handgrip strength and dietary data. A handgrip strength <25th centile of normative reference data has been selected to identify those who might benefit from optimising muscle function[23]. Participants will be provided with study information and consent sought (with up to 1 week to consider participation).

Inclusion criteria:
- Handgrip strength <25th centile
- Currently on a PKU diet
- Taking ≥75% of prescribed protein substitutes weekly

Exclusion criteria:
- Unable to take protein substitutes for an extended period of time

Randomisation and schedule of treatment

Participants will be randomised (with support from CTU) to either one of two intervention arms:
- ***NB protein requirements (standard care)*** *plus* home-based exercise programme
- ***IAAO protein requirements*** *plus* home-based exercise programme

Dietary intervention

Following randomisation, the participant's dietary regimen will be discussed and a supply of additional protein substitutes arranged for the study duration if required.

Exercise intervention

A 12-week evidence-based home RET programme[41,42] will be developed by a physiotherapist (funded into this application) using the rehabilitation treatment specification system (RTSS)[43]. Rehabilitation interventions in research often lack specificity. The RTSS overcomes this by utilising a common language, improving research reporting and allows the refinement of the intervention's underlying treatment theories[44]. One designated physiotherapist, blinded to dietary group allocation, will deliver the exercise intervention. Each participant will receive written guidance, exercise bands, and a one-to-one video induction session to ensure correct exercise technique and intensity.

Monthly video drop-in sessions will be run to offer participants the opportunity to discuss any questions regarding these interventions. There will be two independent sessions, one with the physiotherapist and one with myself, for participants to drop-in as required. Running separate sessions will ensure the physiotherapist remains blinded to the dietary intervention.

Data collection

All measurements taken on the test day of the observational study will form participants' baseline measurements for the feasibility study. During the feasibility study, participants will complete fortnightly dried bloodspots (metabolic control) and the IPAQ-SF (physical activity status). Adherence to the dietary and exercise intervention will be monitored through a weekly log, including the monitoring of exercise duration and intensity.

Participants will return to KCL for a second test day at the end of week 12. Financial compensation will be provided. I will perform all assessments, and data collected will be entered into REDCap EDC.

Additional measurements

Ultrasound – Bicep, forearm and quadriceps muscle layer thickness and rectus femoris cross-sectional area (RF_{CSA}) will be measured by ultrasound at Weeks 0 and 12. These measures of muscle mass closely correlate with DXA[45,46] and MRI[47] and muscle function in chronic disease[48]. RF_{CSA} has demonstrated sensitivity in detecting changes in muscle mass in response to resistance training in COPD[49]. Ultrasound is a novel technique to assess muscle mass in people with PKU, and has advantages in being portable, low-cost and non-invasive, and would also represent a significant extended scope of practice for myself as a Dietitian.

Feasibility measures

The following feasibility parameters will be assessed to inform the intervention development for the full-scale RCT, planned for my post-doctoral research:

Recruitment, randomisation, and attrition – The number of adults with PKU that meet eligibility criteria, the proportion of eligible willing to participate and be randomised, and the retention rate, will be recorded.

Acceptability and practicality of the interventions and measurements – The Acceptability of Intervention Measure and Feasibility Intervention Measure (2-minute questionnaires)[50] will be completed by participants at week 6 and 12. Furthermore, semi-structured online/telephone interviews will be conducted using an interview guide based on TFA at week 12, recorded digitally and transcribed by myself. Dietary and physical activity assessments will evaluate whether participants can successfully implement and sustain the dietary regimes and RET programme. The acceptability and practicality of implementing the interventions in a healthcare service will be explored further through a focus group with Dietitians providing the dietary support to this patient group.

Outcome parameters: The sensitivity and specificity of common clinical measurements *vs*. gold standard measurements will be established, including their mean (SD) of absolute and change values and anticipated effect sizes; therefore, determining the measurements to be used in the full-scale trial and informing the sample size calculation.

Data Analysis

Sample size

A formal power calculation is not required for a feasibility study. However, an adequate sample size will ensure the above feasibility measures can be determined. Twelve participants per group have been previously recommended for such studies[51], and therefore this study will aim to recruit 24 participants in total. With 38% of adults with PKU previously found to have a grip strength <3rd percentile[19], it is anticipated that 24 participants can be recruited from Study 1 (n=50) with inclusion criteria of a grip strength <25th percentile.

Analysis

Effectiveness conclusions will not be drawn from this feasibility study, but exploratory comparative analysis will be performed on endpoint data. Feasibility measures will be reported descriptively, including numbers eligible, consented, randomised, completed, and also data completeness. All analysis will be conducted using SPSS Statistics. All analyses will be performed on an intention-to-treat and per protocol basis. Analysis of the interviews and focus group will use the five stage Framework Analysis methodology to deductively code and organise data according to the TFA[40,52].

6. Dissemination, Outputs & Impact

Research outputs

- **Conference presentation:** I will submit research abstracts to relevant national and international metabolic conferences.
- **Publications:** I plan to publish the findings of my research in high impact peer-reviewed journals. There will be at least two publications, one from the observational study and one from the feasibility study. In addition, there will be method development publications relating to improving clinical assessment in PKU (e.g. US vs DXA, FFQ vs food diary). All will be published open access to ensure wide dissemination.
- **Healthcare innovation:** My research will help to establish reliable markers of protein status to incorporate into the clinical assessment and monitoring of patients with PKU. This will advance current clinical practice in determining dietary requirements and monitoring the outcomes of dietary interventions.

Research dissemination to patients and professionals

I am keen to disseminate my findings to the patients who may benefit from it. Our PPIE group will be actively involved in disseminating this research. During the first year of my Fellowship, I will work with our PKU PPIE group to develop plans for informing service users and other NHS services nationwide of the research outputs, and engage funding bodies to support future research.

The National Society of PKU (NSPKU, patient support charity) are key stakeholders in disseminating research to national and international PKU communities via their website, social media and annual conference. I will collaborate with NSPKU to disseminate my research via these platforms.

The BIMDG specialist group consists of clinicians, clinical service leads and national advisors. I plan to present my research at BIMDG conferences and group meetings to inform and engage key stakeholders influencing NHS service provision for patients with PKU.

Impact on the health system (guidelines)

I intend to contribute to the review and development of dietary management guidelines informing clinical practice in the UK and Europe, such as in the ESPKU guidelines (last published 2017)[9] and the UK consensus on Dietetic Management of Adults with PKU (published 2022)[3]. These guidelines are reviewed every 3-5 years. My research will contribute to the evidence informing protein requirement guidelines, dietary interventions, and assessment and monitoring of protein status in adults with PKU.

Further research funding

I plan to develop a proposal for post-doctoral research through the NIHR ACAF scheme to undertake a full-scale

complex intervention trial to investigate the impact of dietary regimens and lifestyle factors to optimise muscle mass and function in adults with PKU. In addition, other NIHR and MRC research grants will be highly relevant to my research area and will be targeted for future research funding.

Barriers to further research

An ongoing barrier to research in PKU is subject numbers due to the rare nature of the condition. However, 25% of patients within our service have PKU, and with our patient numbers doubling every 5 years, I anticipate there will be adequate numbers in our service alone to enable a large RCT.

Research Impact

Short-term (within 3 years): I plan to establish new care pathways for PKU clinical assessment. Incorporating novel anthropometric and functional assessments will extend the scope of dietetic practice and allow patients to receive more individualised care.

Medium-term (within 3-5 years): Outcomes of my research will inform the development of guidelines (in dietary assessment, monitoring and intervention) that will in turn inform care pathways and practice standards for PKU services nationally and internationally. Therefore, clinical practice will evolve to ensure optimal health outcomes for PKU patients as they age. Furthermore, the feasibility study will inform the design of a full-scale clinical trial to further investigate diet and lifestyle factors that modulate muscle mass and function in adults with PKU.

Long-term (within 5+ years): With the revised guidelines, dietary assessment and monitoring of muscle mass and function will be incorporated into routine clinical practice. With standardised assessment and monitoring, multi-centre data can be collated, providing novel insight into the impact of PKU on health outcomes with ageing. Outcomes from the full-scale clinical trial will inform further research into dietary advances to support ageing well with PKU.

7. Project Management

The Project Advisory Group will consist of myself, my academic and practice supervisors, physiotherapist and PPIE members. The group will meet at the beginning, mid-point and end of each study to discuss study implementation, recruitment, retention, any issues raised during the study and progression against the agreed timetable. In addition, the six core elements of the MRC/NIHR framework will be considered by the group to inform how the research should proceed[39]. These meetings will be via video. Financial reimbursement has been budgeted for three PPIE members' involvement in these meetings. A diverse PPIE will be selected to ensure members represent the views of the wider PKU community.

I will meet quarterly with KCL Finance for support with financial management of the Award.

8. Ethical considerations

The majority of measures and interventions used in this study are widely used in other areas of the NHS or in research. However, I have worked with the PPIE group to identify whether specific aspects of the study may have ethical concerns, and to discuss how to mitigate them, as follows:

i. Some participants may feel sensitive about the results of their body composition and functional assessments and wish to have results withheld. This option will be offered to participants.
ii. The venous blood samples and bloodspots can cause brief pain. These tests are part of routine care in PKU and will therefore be tolerated by most participants. Additional guidance on bloodspot technique will be offered to participants.
iii. All biochemical results will be reviewed by Dr Ramachandran (Supervisor and Metabolic Service Lead GSTT), with any abnormal results being reported and referred to appropriate services.
iv. Participants identified through the observational study who find dietary adherence challenging (i.e. taking <75% of protein substitutes), will be offered a clinic follow-up review with a Dietitian in the GSTT service to support them.

Participants have the right to refuse participation or withdraw from the studies without giving reasons and without

prejudicing their ongoing care.

9. Success Criteria

Observational study: Complete data set from at least 50 eligible participants

Obtaining adequate numbers of eligible participants is achievable from within the patient cohort at GSTT. However, the service covers a large geographic area and there is a risk that some participants may not wish to travel to central London. To overcome this, compensation for travel will be provided and, if required, the test day will be booked for the same day as the patient's routine clinic appointment.

Feasibility study: Recruitment of 24 eligible participants and adequate retention to establish feasibility of progression to full trial

The eligibility criteria for the feasibility study were selected to identify participants with compromised handgrip strength. If <24 participants are eligible, this is an important feasibility outcome, and the handgrip strength cut-off will be revised to ensure an adequate sample size for the feasibility study.

Our PPIE group have provided strategies to support retention during the feasibility study, such as organising virtual drop-in sessions.

A traffic light system, as recommended in best practice guidelines[53], provides criteria that will be used to guide progression of the feasibility study to a full-scale trial (see Table 1).

7. Patient & Public Involvement

Please describe how patients/service users, carers and the public have been involved in developing this proposal

Phenylketonuria PPIE group

I founded the first Phenylketonuria (PKU) PPIE group at GSTT. This group has been pivotal in identifying research priorities and informing this research proposal. In our first meeting in 2021, we discussed research priorities, identifying that adults with PKU had concerns about the impact of PKU and its dietary management on long-term health outcomes and functional ability in later life. This, along with my narrative review[1], scoping review[2] and my nationwide survey of patients[5], led to the development of the research questions. A recent PPIE meeting to revisit the research proposal (based on feedback from my initial NIHR DCAF application), confirmed that this remains a considerable research priority.

The research design evolved during multiple meetings with this PPIE group who informed the study design as follows:

- Recruitment methods (suggesting screening and consent via video)
- Increasing test day attendance (offering compensation, written letters to request time off work and provision of a suitable meal)
- Test day assessment methods were revised based on members' feedback. The PPIE group raised concerns about some participants feeling self-conscious about their body composition and functional assessment results and provided suggestions to overcome this potential issue (see 'Ethics')
- Ideas of written communication for the feasibility study, including a fridge calendar and journal to track progress, and drop-in video sessions for participants to discuss any questions about the diet and exercise interventions
- Reviewing of Plain English Summary (PES)

The Research Support Service also reviewed and provided feedback, as a non-PKU audience, on the PES and research proposal.

Wider adult PKU community

I conceived, designed, conducted, analysed and published a nationwide questionnaire survey to examine the knowledge, perceptions and dietary behaviours of adults with PKU[5]. Findings confirmed the importance of this research in adults with PKU, with >50% having concerns for their long-term health, that are currently being unmet. Up to 75% of people with PKU considered research focused on the impact of the PKU diet on muscle and health and how regimens can be tailored to body composition and physical activity status, as either *extremely important* or *very important*, highlighting the value of this research.

Please describe the ways in which patients/service users, carers and the public will be actively involved in the proposed research, including any training and support provided

Phenylketonuria PPIE group

I will continue my partnership with the Phenylketonuria (PKU) PPIE group at GSTT, who will be consulted regularly throughout the fellowship.

This group brings lived experience, contributing valuable advice on engaging the PKU community to optimise participation in research and ensuring that research meets the needs of this cohort.

The PKU PPIE group will be involved in, and advise on, the following during my fellowship:

- Advertising materials and methods to optimise recruitment
- Patient information sheet and consent forms to ensure clarity in the study and lay terminology so participants can make an informed choice about participation
- Questionnaire and semi-structured interview design: questionnaires and interview questions designed by the research team will be piloted with members of the PPIE group to test clarity and interpretation of questions, readability, length and interview structure
- Developing participant information resources for the feasibility study (eg. the exercise guide, fridge

calendar and the journal). Members will advise on the design, layout and wording used for these resources
- Dissemination methods for communicating this research to the PKU community and other IMD services
- Planning for further research (eg. post-doctoral funding)

Project Advisory Group

Three PPIE members will join the Project Advisory Group to support discussions on study implementation, recruitment, retention and any issues raised during the study (see 'Research Plan'). A diverse PPIE group will be selected to ensure members can represent the views of the wider PKU community.

Supporting, enabling and capturing participation

Meetings will be held online to enable involvement from a wide geographical area. Financial reimbursement for the above consultations has been costed into this fellowship to compensate for PPIEs' time and active involvement, calculated using the NIHR payment guidance. New members to the group will be offered training in PPIE through NIHR Biomedical Research Centre at GSTT (or similar) to ensure they feel equipped and confident to contribute to these meetings. Furthermore, new members will be buddied with an established member for support and mentoring.

The involvement and impact of the PPIE on this research and informing future research will be captured and published utilising the GRIPP 2 reporting checklist[54].

If it is considered not appropriate and meaningful to actively involve patients/service users, carers and the public in your proposed research, please justify why
Not applicable

8. Training & Development and Research Support

Proposed training and development programme

I have designed a training programme to develop both my academic and clinical skills in consultation with my supervisors and using the Vitae Researcher Development Framework (see 'Uploads' for training timetable).

Formal study

The Department of Nutritional Sciences at King's College London (KCL) fosters development of its doctoral students through the Health Sciences Doctoral Training Centre (HSDTC) and the King's Clinical Academic Training Office (KCATO). Through the HSDTC and KCATO, I have selected courses in critical appraisal, managing research data, academic writing, preparing for the upgrade, thesis writing, preparing for the viva, and clinical academic career development courses.

Academic Training and Development

Research design and methods

Qualitative methods: I am keen to develop and strengthen my skills in qualitative research, especially for the feasibility study and future research, and will attend the following courses offered by the Social Research Association and National Centre for Social Research:

- Introduction to Qualitative Interviewing: This 1-day course introduces the method of qualitative interviewing including effective interviewing, and key ethical and practical issues for consideration.
- Introduction to Focus Groups: A 1-day course to develop skills in conducting focus groups.
- Questionnaire design: This 2-day course provides hands-on experience of designing questionnaires and survey questions.

Clinical trials: I intend to advance the feasibility study to a full-scale trial in post-doctoral research, in addition to leading further research in adults with IMDs, including complex interventions. I will attend two courses on clinical trials:

- 'Randomised Controlled Trials Course: A Guide to Design, Conduct, Analysis, Interpretation and Reporting' (University of Oxford). This 5-day course covers fundamental components of planning, conducting and completing successful RCTs.
- 'Developing and Evaluating Complex Interventions' (University of Southampton): A 3-day course structured around the MRC/NIHR Framework for developing and evaluating complex interventions.

Data analysis and interpretation (quantitative and qualitative)

To enhance my skills in data analysis and interpretation for both quantitative and qualitative research, I will attend the following courses:

- 'Practical Statistics for Medical Research' (University College London): This 4-day course teaches essential elements of research methodology and statistics, with a focus on understanding appropriate methodology and interpretation of results.
- Advanced statistical analysis courses using SPSS and R (KCL), teaching the practical aspects of performing more complex analysis using specific software (e.g. linear and logistic regression analyses for my observational study).
- Qualitative Data Analysis (Social Research Association): A 1-day online course equipping researchers with skills to undertake analysis of qualitative data.

Research Governance

I will attend Informed Consent Training / Principal Investigators Essentials / Good Clinical Practice training: These courses are provided free by the NIHR and Biomedical Research Centre (BRC).

PPIE

I will undertake further PPIE training with attending courses provided by NIHR, including NIHR BRC Cambridge PPIE online training sessions on facilitating successful PPIE groups, using PPIE to communicate research and evaluating PPIE.

Dissemination

Nature Masterclass in Scientific Writing and Publishing: Through KCL, I will have access to these Masterclasses, and plan to complete training on scientific writing, publishing, effective communication in research, and peer review.

Overseas visit

I will spend a week with Professor Luc van Loon (see 'Collaborations') at Maastricht University, to observe clinical trials in protein metabolism, and to acquire knowledge in research design and advanced methods of investigating protein metabolism.

Academic and clinical nexus

The following training will provide me with the skills to undertake the assessments included in my research proposal, which will also be incorporated into my clinical practice, extending my scope of dietetic practice.

- Level 1 ISAK Anthropometry Course (St Mary's University): 3-day course to develop my skills in anthropometric profiling including skinfold analysis.
- Functional assessments: Under the supervision of Dr Oliver Witard, Exercise Physiologist, I will develop my skills in conducting laboratory assessments of muscle functional capacity and establish their application to clinical practice.
- Ultrasound: Dr Danielle Bear, Consultant Dietitian (GSTT), a former NIHR ICA awardee has extensive experience in ultrasound techniques for assessing skeletal muscle mass, including publishing on this as part of her NIHR fellowship. Under the guidance and supervision of Dr Bear, I will develop my skills in conducting ultrasound assessments.
- Ionising Radiation (Medical Exposure) Regulations (IR(ME)R) certification: A 1-day course that is a pre-requisite to undertaking DXA assessments.
- Phlebotomy skills 2-day course.

Clinical research integration visit: I will spend a week with Professor Anita MacDonald (see 'Collaborations') at the Birmingham Children's Hospital to observe research in IMD and how their research translates into clinical practice, and to discuss dietary management priorities for patients with IMDs.

Leadership: Through the NIHR, I will have access to the Leadership e-learning modules. I plan to complete these to develop and strengthen my leadership skills across both my academic and clinical roles.

Professional Training and Development

Clinical mentorship

As the lead Senior Specialist Dietitian, I will continue to provide regular mentorship to the Dietetic team. This continuity in my involvement with the clinical team will allow me to maintain my clinical skills and translate my research into clinical practice.

Conference attendance

Attending, and contributing to, relevant national and international metabolic conferences will ensure my clinical knowledge remains current.

Clinical practice

Care Innovation

Along with the professional development training outlined above, I will establish new care pathways for PKU clinical assessment, adopting the skills gained in anthropometric and functional assessments. These assessment techniques are novel in IMD and would allow for more individualised assessment and tailored dietary interventions. Maintaining my involvement in these clinics will ensure that these developments translate into improved patient care.

With the NHS advancing use of virtual patient consults, GSTT IMD service has adopted patient webinars to deliver updates in clinical management. I will contribute to the development, updating and running of webinars and provide specialist guidance for complex questions regarding dietary management of IMDs.

International Clinical Advisor

I will continue my role on the Advisory Board for the Global DARE Foundation including annual revision and publication of dietary guidance, hosting training webinars, and providing ongoing clinical expertise to Dietitians internationally.

Clinical educator

I will continue to provide clinical education on the dietary management of IMD for dietetic students at KCL (maximum 6-hours/year).

Primary Doctoral Supervisor

PRIMARY DOCTORAL SUPERVISOR 1

Name	Professor Kevin Whelan

Institution	King's College London

Position	Professor of Dietetics and Head of Department of Nutritional Sciences

Description of support provided
Professor Whelan will be my Primary Doctoral Supervisor. He is the inaugural Professor of Dietetics at King's, and a Fellow of the British Dietetic Association. He is a world expert in translational dietetic research who has led numerous observational studies, feasibility studies and clinical trials to understand the application of nutrition to improve clinical outcomes and quality of life in patients with chronic diseases, aligning with the focus of my proposed research. He has expertise in data analysis (including teaching Statistics to Masters students) and will provide me with statistical support, expertise that will be invaluable to my PhD fellowship. Professor Whelan has published >220 full papers in peer-reviewed journals, including on body composition, dietary intake and patient reported outcomes. He has received funding >£7.8 million, and in 2021 was a Clarivate Highly Cited Researcher for multiple highly cited papers in the top 1% for field. Professor Whelan has supervised 16 PhDs to completion, including two previous and one current NIHR doctoral fellowships. Professor Whelan is a committed educator with an MA in Higher Education Teaching and was awarded the King's PhD Supervisory Excellence Award 2021. Professor Whelan is a HEE/NIHR Mentor, and former member of the HEE/NIHR ICA Clinical Lectureships Awards Panel. He supervised my NIHR PCAF, therefore we have already collaborated on two research studies, and published three papers together on PKU, protein and muscle. We have an established and positive working relationship and Professor Whelan has contributed significantly to supporting my research training and developing this proposal.

Additional Academic Supervisor(s)

ADDITIONAL ACADEMIC SUPERVISOR 1

Name	Dr Oliver Witard

Institution	King's College London

Position	Senior Lecturer in Exercise Metabolism & Nutrition

Description of support provided
Dr Oliver Witard will be my Academic Supervisor. His research explores the interaction of exercise and nutritional interventions, primarily protein nutrition, to offset the age-related decline in muscle mass and function. His expertise is therefore directly aligned with my research proposal. Dr Witard has supervised five PhD students to completion (three as primary and two as secondary supervisor), and is currently supervising one PhD as primary, and another as secondary, supervisor. During my NIHR PCAF, I collaborated with Dr Witard on several research activities, culminating in the publications of a narrative review and scoping review on protein requirements and protein status in people with PKU. Dr Witard has been involved throughout in developing this doctoral research proposal, providing vital knowledge and insight to the research design and the proposed functional measurements. Dr Witard has extensive experience in the management of human trials in protein nutrition and muscle metabolism research, including the assessments of muscle functional capacity. He has led on multiple studies investigating nutritional and exercise interventions in young and older adults with the primary outcome targeted at optimising skeletal muscle mass and function. His expertise will be invaluable during my fellowship and by informing the design of further research. Dr Witard will provide training and supervision in conducting the functional assessments as part of the observational and feasibility studies. Under the supervision of Dr Witard, I will have access to state-of-the-art Exercise Physiology laboratories in the Centre for Human and Applied Physiological Sciences where my research will be conducted.

Clinical Supervisor(s)

PRACTICE SUPERVISOR 1

Name	Dr Radha Ramachandran

Institution	Guy's and St Thomas' NHS Foundation Trust

Position	Service Lead and Consultant in Adult Inherited Metabolic Diseases

Description of support provided
Dr Radha Ramachandran will be my Practice and 3rd Academic Supervisor. She was appointed Consultant in Metabolic Medicine in 2013. Dr Ramachandran chairs the UK Adult Inherited Metabolic Diseases (IMD) Physicians' Group and sits on the British IMD Group Clinical Guidelines and NICE Diagnostic Accuracy Committees. She has international expertise in, and sits on Advisory Boards for, ORPHAN lipid and peroxisomal

disorders.

Dr Ramachandran completed an NIHR Doctoral Research Fellowship in 2011 during specialist training in Chemical Pathology and Metabolic Medicine at Imperial College Healthcare NHS Trust. Pursuing her research interests, she has successfully supervised BSc and MSc projects, is Principal/Co-investigator for commercial Phase 1, 2 and 3 trials for rare IMDs, and has supervised my NIHR PCAF.

I work closely with Dr Ramachandran to jointly manage IMD patients with complex nutritional needs. We interact daily in multidisciplinary clinics, and in departmental service development and research meetings. She has provided continued clinical supervision and, as Service Lead, opportunities to advance the dietetic service - both vital for my professional development as Lead IMD Dietitian. We have collaborated on several audits, service improvements and research projects, leading to a number of abstracts, international presentations and publications.

Given the clinical mentorship she has provided thus far, our shared clinical and research interests and the need for a multidisciplinary approach to managing patients with IMD, having Dr Ramachandran as my Practice and Academic Supervisor will provide me with the support needed to further develop my career as a clinical academic in IMD.

PRACTICE SUPERVISOR 2

Name	Dr Danielle Bear

Institution	Guy’s and St Thomas’ NHS Foundation Trust and King’s College London

Position	Consultant Dietitian – Critical Care (GSTT) and Honorary/Adjunct Lecturer (KCL)

Description of support provided

Dr Danielle Bear is a Consultant Dietitian recognised nationally and internationally for her expertise in the nutrition management of critical ill patients. Dr Bear has been awarded British Dietetic Association Roll of Honour three times, and is a past recipient of the Rose Simmonds and Elizabeth Washington awards.

Dr Bear established the critical care Dietetic service at GSTT and has extensive experience in leadership and fostering professional development. Over the last seven years, Dr Bear has provided mentorship supporting my professional development and research experience, and her ongoing support will be invaluable as I further develop and expand my professional skillset.

Dr Bear embarked on her career as a clinical academic following completion of an NIHR CDRF in 2020. Her doctoral research investigated the measurement and prevention of skeletal muscle wasting in critical illness. Dr Bear has continued research exploring nutrition interventions, including protein requirements, to reduce muscle wasting, and is an expert in techniques to measure body composition in clinical practice. Her expertise is therefore directly aligned with my research proposal, and will be invaluable during my fellowship and the implementation of research findings into clinical practice.

Dr Bear bridges both clinical and academic settings in her role as a Consultant Dietitian at GSTT and honorary lecture at KCL. She has supported numerous Dietitians to undertake further academic training and research, including supervising MSc students. Dr Bear brings experience of developing a clinical academic career in Dietetics, which will be instrumental in supporting me in developing my career.

Collaborations

In preparing this application, I established the following collaborations that I will continue to develop to support my research and training.

Centre for Human and Applied Physiological Sciences, KCL

The Centre is committed to integrative and translational research focused on human physiological function and adaptation in health and disease. Based on REF 2021, CHAPS is a leading UK centre (2nd in UK) with 75% of publications ranked as world leading. Through my collaboration with Dr Witard (Supervisor), I will have access to CHAPS facilities and the physiologists/exercise scientists with expertise in muscle, ageing and exercise. Developing collaborations with the scientists will strengthen the partnerships between KCL and the GSTT Adult Metabolic service to support my doctoral, and future, research.

Physiotherapy Department, GSTT

Dr Jacky Jones, Deputy Chief AHP Research Lead and Head of Physiotherapy, and clinical physiotherapists at GSTT, informed the proposal for the feasibility study exercise intervention. I will continue to collaborate with Dr Jones and a physiotherapist during my fellowship on the design, implementation and analysis of the exercise intervention. This will strengthen my partnership with the Physiotherapy Department, who will be integral to future IMD research involving exercise interventions.

Professor Anita MacDonald, *Consultant Dietitian, Birmingham Children's Hospital.*

Professor MacDonald is an international leader in the management of IMDs and has authored over 200 publications, many in PKU research. I worked with Professor MacDonald in a multi-centre study and visited her team to observe clinical research in PKU. I will further develop this collaboration with Professor MacDonald and her research team during my fellowship. Professor MacDonald's experience in translational research in PKU and clinical expertise will be invaluable in fostering my development as a clinical academic. She is a clinical academic that I would like to emulate, with my career ambition to be a Clinical Professor of Dietetics in IMDs.

International: Professor Luc van Loon, *School of Nutrition and Translational Research in Metabolism, Maastricht University Medical Centre+ (MUMC+)*

Professor van Loon is the scientific coordinator of the Stable Isotope Research Centre at MUMC+ and has world-leading expertise in protein metabolism research and tracer techniques. Establishing collaborations with Professor van Loon and his research team will provide me with the opportunity to develop skills in the application of tracer techniques to apply in future research investigating protein requirements for people with PKU.

Patient organisations

The National Society of PKU (NSPKU, patient support charity) has a long-standing involvement in PKU research and a large patient and healthcare professional membership. I have developed collaborations with the NSPKU through my clinical role and will continue to foster this partnership during my fellowship and post-award, providing opportunities for me to engage with key stakeholders and service users from across the UK in research development, participation and dissemination.

Industry: Arla Foods Ingredients (AFI)

Through Dr Witard's (Supervisor) longstanding collaboration with AFI, I have developed links with the AFI nutrition scientists. AFI has an interest in protein metabolism and protein substitute development for PKU, and has funded multiple studies in this area. This collaboration will be important for further research collaborations and in supporting the translation of the research outcomes into patient care. I plan to continue to strengthen this industry collaboration during my fellowship and post-award.

Healthcare professional groups

I will continue my collaborations with clinicians and academics working in IMD through my involvement in national and international IMD groups. Outcomes from my doctoral research will influence patient care and management guidelines, and therefore, collaborations with these group will be important to disseminate findings, contribute to guidelines and collaborate on future research.

Host Organisation Support Statement

The following statement of support should be composed jointly by the Heads of Department / Senior Managers of the employing host organisation and the partner organisation.
The Heads of Department / Senior Managers of these organisations will be able to add the statement of support to

the form once they have confirmed participation. Invitations to participate are sent by the applicant via the 'Participants and Signatories' section of the form.

Statement of support

Sarah Firman plans to complete her ICA fellowship at King's College London (KCL, Host Organisation) and Guy's & St Thomas' NHS Foundation Trust (GSTT, Partner Organisation). Both are founding institutions of the King's Health Partners (KHP) Academic Health Sciences Centre, whose aim is to unite world-class research, education and clinical practice and translate this into excellent patient care, therefore providing the perfect organisational support for Sarah's Fellowship.

Sarah is the Lead Senior Dietitian in the Inherited Metabolic Diseases service at GSTT and has established the service over the last seven years. During this time, she has demonstrated drive to enhance patient care through research, development of care pathways and extending scope of dietetic practice. Sarah was successfully awarded an NIHR PCAF in 2019, and presented her PCAF research at conferences, it was discussed in national meetings, and she has published all four outputs from it. There are existing, supportive and productive research collaborations between Sarah and all her academic and clinical supervisors that will be strengthened during this DCAF.

The Department of Nutritional Sciences at KCL was the first university department of its kind in Europe and is recognised as a world-leader in Nutrition and Dietetics research. In REF 2021, the Department was returned under Unit of Assessment 3 (Allied Health Professions, Dentistry, Nursing and Pharmacy) where 70% of research was 4* (world-leading) and 95% of research was 4* and 3* (world-leading or internationally excellent). This resulted in the highest ranking in the UK for UoA3 for research, research output, research GPA and research power. Importantly for Sarah's PhD, 100% of the research environment was rated 4* (world-leading) and KCL has one of the highest PhD completion rates in the UK.

The Department of Nutritional Sciences at KCL has a well-developed infrastructure to support and monitor the development of postgraduate researchers. Sarah will have access to the largest collection of nutrition journals in the UK. She will be provided with two academic supervisors, a combined academic & clinical supervisor, and another clinical supervisor, a thesis progression committee, an induction programme, regular monitoring of progress, and development of research skills specific to her fellowship. There is a well-equipped Metabolic Research Unit for clinical research visits, the Centre for Human and Applied Physiological Sciences at KCL for function assessments, and our Statistical Support Unit for Sarah's academic training and development throughout her DCAF.

GSTT demonstrates a commitment to progressive academic study and supporting clinical academic roles for Allied Health Professionals (AHPs). Dr Jacky Jones has just been appointed as the first AHP Research Lead for the organisation and has a position on the Trust R&D board. The Trust has recently published their AHP strategy with research as a stated priority, including supporting clinical academic careers, hosting of clinical academic posts and commitment to supporting clinical academic training.

The Department of Nutrition and Dietetics at GSTT is the largest Dietetic Department in the UK with dietetic leaders in several areas of clinical care, and extensive experience supporting AHP clinical researchers. Our Trust has four Consultant Dietitians (the largest number of any Trust in the UK) all with PhDs (including two funded by NIHR awards), and a number of other staff with PhDs and MRes.

There is a strong history of successful support of clinical academic careers among dietitians across the departments at King's Health Partners (including KCL and GSTT).

There are currently three joint-funded permanent clinical academic dietitians (the largest number of Clinical Academic Dietitians at any HEI in the UK) who lead clinical services at GSTT and lead research at KCL. There have been two former HEE/NIHR Lectureships at King's Health Partners both of whom successfully completed projects and moved to clinical academic positions elsewhere. There are three previous and one current recipient of HEE/NIHR ICA Doctoral Fellowships all of whom successfully completed their PhDs on time and are now in Clinical Academic positions. Across King's Health Partners, our combined departments represent the largest group of academic, research and clinical academic dietitians in the UK.

Creating an inclusive environment that puts colleagues, researchers and patients first, and which promotes equality, diversity, and inclusion (EDI) are key priorities for both KCL and GSTT. In 2021, KCL were awarded the institutional Athena Swan Silver accreditation recognising their work to advance gender equality. The Faculty, School and Department in which Sarah will be based have EDI Leads who report into College council, as well as an EDI forum for students and staff. GSTT has an incredibly diverse workforce and takes proactive measures at all levels to make EDI a part of everyday business. Further details of approaches to EDI are available from KCL and

GSTT.

Sarah has chosen to be seconded to KCL and retain substantive employment with GSTT for the duration of her fellowship. This will provide Sarah with the opportunity to be integrated into the Department of Nutritional Sciences at KCL, where she will undertake her research, working alongside over 20 doctoral students, while remaining connected to the clinical/patient setting. The Department of Nutrition and Dietetics at GSTT supported Sarah by backfilling her post during her HEE/NIHR PCAF, and will do so again should she be successful.

Following her PhD, we are committed to supporting Sarah in her ambition to become a leading clinical academic in the dietary management of Inherited Metabolic Disorders and we will work together to build a business case for a joint clinical academic appointment. Of note, all of our previous dietitians who received an HEE/NIHR ICA Doctoral Fellowship were subsequently appointed as Consultant Dietitians at GSTT and Clinical Academics at KCL through permanent joint funding, and we anticipate the same for Sarah.

We emphatically support Sarah in this doctoral fellowship application and in her developing an integrated clinical academic career.

Rhys White MRes RD, Guy's & St Thomas' NHS Foundation Trust
Professor Kevin Whelan PhD RD FBDA, King's College London

9. Detailed Budget

The finance section should provide a breakdown of costs associated with undertaking the research as described in the proposal.

1

Staff type	Lead Applicant

Name	Sarah Firman

Employing organisation	Guy's and St Thomas' NHS Foundation Trust

Grade	7

Starting point on scale	8

Increment Date	30/05/2025

Justification of Costs
This is the salary for the lead applicant for the 36 month Fellowship period.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	50056
Geographical Weighting (£)	7746
Other allowances (£)	289
Superannuation (£)	7330
Nat. Insurance (£)	8312
Current Annual Costs (£)	**73733**

2

Staff type	Shared Staff

Name	Physiotherapist

Employing organisation	Guy's and St Thomas' NHS Foundation Trust

Role on this research	Physiotherapist

Grade	7

Starting point on scale	8

Increment Date	

Justification of Costs
Input from a trained physiotherapist would be required to ensure the design for the exercise intervention integrates both the correct dose and intensity for the participants and the safe implementation thereof. The specialist physiotherapist will be involved in the feasibility study (Year 1 and 2) and will be responsible for: • Developing the home-based resistance exercise programme • Providing a one-to-one induction session for each participant • Attending the monthly drop-in video sessions for participants to ask any questions about the exercise interventions • Participating in the Project Advisory Group meetings to finalise exercise intervention and implementation, and discuss study progression and outcomes.

Details of Posts and Salaries Costs

	Total (£)
Salary (£)	5006
Geographical Weighting (£)	775
Other allowances (£)	29
Superannuation (£)	831
Nat. Insurance (£)	733
Current Annual Costs (£)	**7374**

Direct Costs
These are the costs that are specific to the research, which will be charged as the amount actually spent and can be supported by an audit record.

Staff Type: (Lead Applicant)-Mrs Sarah Firman
Type of cost: NHS
% Full-time on this research: 100.0
Total months on this research: 36.0
Justification of Cost: This is the cost to cover the salary for the Lead Applicant for the 36 month Fellowship period. The type of cost selected is 'NHS' as the Lead Applicant will be employed by an NHS Trust.

	Year 1	**Year 2**	**Year 3**	**Total Cost (£)**
Basic Salary	50056	50056	50056	150168
National Insurance	7330	7330	7330	21990
Superannuation	8312	8312	8312	24936
Geographical weighting / Other allowances (if applicable)	8035	8035	8035	24105

Staff Type: Other
Other Type: Shared Staff
Name: **Physiotherapist**
Type of cost: NHS
% Full-time on this research: 10.0
Total months on this research: 24.0
Justification of Cost: The specialist physiotherapist will be responsible for developing and implementing the exercise intervention as part of the feasibility study. This intervention will require expertise to ensure the exercise programme includes the appropriate dose and intensity and to ensure participants can undertake the exercises safely using the correct techniques. The physiotherapist will also support participants during the study with monthly drop in sessions and participate in the Project Advisory Group meetings. The cost covers 0.1 WTE for Year 1 and 2. The type of cost selected is 'NHS' as the physiotherapist will be employed by an NHS Trust.

	Year 1	Year 2	Year 3	Total Cost (£)
Basic Salary	7374	7374	0	14748

Totals	Applicant	Support Staff	Shared Staff	Total Cost (£)
Total HEI	0	0	0	0
Total NHS	221199	0	14748	235947
Total Commercial	0	0	0	0
Total Other	0	0	0	0

Travel, Subsistence and Conference Fees

Type of cost: HEI
Description: Conference Fees
Conference: True
Item Description: Conference registration, travel and accommodation
Justification of Cost: Please see justification for conference attendance under 'Training and Development' and 'Dissemination'. These fees are to cover the registration fees, travel and accommodation to attend the annual British Inherited Metabolic Diseases (BIMDG) national conference and an annual international conference: Society for the Study of Inborn Errors of Metabolism (SSIEM) or the International Congress of Inborn Errors of Metabolism (ICIEM). The SSIEM/ICIEM are the largest inherited metabolic diseases conferences internationally. I have regularly attended the BIMDG and SSIEM/ICIEM conferences over the last 6 years and presented several research abstracts. Attending these conferences will ensure my clinical knowledge remains current, provide an opportunity to develop collaborations and be a means to disseminate my research findings.

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	1000	1000	3000

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Journey costs for overseas visit (Maastricht University)
Justification of Cost: The justification for this overseas research visit is outlined in the 'Training and Development' and 'Collaborations' sections. This cost is to cover return flights from London to Amsterdam, and a return train journey from Amsterdam to Maastricht. The travel quote was obtained from Guy's and St Thomas' NHS Foundation Trust's preferred travel arrangement company, Clarity Travel.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	376	376

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Accommodation for overseas visit (Maastricht University)
Justification of Cost: Please see justification for this overseas research visit in the 'Training and Development'

and 'Collaborations' sections. This cost is to cover basic accommodation (5 nights) in Maastricht. Quote obtained from booking.com

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	590	590

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Journey costs for site visit (Birmingham Children's Hospital)
Justification of Cost: The justification for this site visit is outlined in the 'Training and Development' and 'Collaborations' sections. This cost is to cover the super off-peak return train journey to Birmingham and is based on the current price provided by www.thetrainline.com

	Year 1	Year 2	Year 3	Total Cost (£)
	0	36	0	36

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Accommodation for site visit (Birmingham Children's Hospital)
Justification of Cost: Please see justification for this site visit in the 'Training and Development' and 'Collaborations' sections. This cost is to cover basic accommodation (4 nights) in Birmingham. Quote obtained from booking.com

	Year 1	Year 2	Year 3	Total Cost (£)
	0	390	0	390

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Journey costs to Oxford for RCT course
Justification of Cost: The justification for undertaking this course is outlined in the 'Training and Development' section. This cost is to cover the return train journey to Oxford and is based on the current price provided by www.thetrainline.com.

	Year 1	Year 2	Year 3	Total Cost (£)
	35	0	0	35

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Accommodation for Oxford RCT course
Justification of Cost: The justification for this course is outlined in the 'Training and Development' section. The course provides accommodation and breakfast at Merton College for £120 per night (five nights).

	Year 1	Year 2	Year 3	Total Cost (£)
	600	0	0	600

Type of cost: HEI
Description: Journey Costs
Conference: False
Item Description: Journey costs to Southampton University for Developing and Evaluating Complex Interventions course
Justification of Cost: The justification for undertaking this course is outlined in the 'Training and Development' section. This cost is to cover the return train journey from London to Southampton Central and is based on the current price provided bywww.thetrainline.com.

	Year 1	Year 2	Year 3	Total Cost (£)
	25	0	0	25

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Accommodation for Southampton Developing and Evaluating Complex Interventions course
Justification of Cost: Please see justification for this course in the 'Training and Development' section. This cost is to cover basic accommodation (3 nights) in Southampton. Quote obtained from booking.com

	Year 1	Year 2	Year 3	Total Cost (£)
	359	0	0	359

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Meals associated with overseas visit (Maastricht University)
Justification of Cost: Please see justification for this overseas research visit in the 'Training and Development' and 'Collaborations' sections. This cost is to cover meal allowance (as per King's College London travel policy) for the 6 days associated with the overseas visit to Maastricht University.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	150	150

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Meals associated with site visit (Birmingham Children's Hospital)
Justification of Cost: Please see justification for this site visit in the 'Training and Development' and 'Collaborations' sections. This cost is to cover meal allowance (as per King's College London travel policy) for the 5 days associated with the site visit to Birmingham Children's Hospital.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	125	0	125

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Meals associated with Oxford RCT course
Justification of Cost: Please see justification for this course in the 'Training and Development' section. As part of the course registration package, daily lunch and one dinner are provided. This cost is to cover dinner allowance for the other 4 nights (as per King's College London travel policy).

	Year 1	Year 2	Year 3	Total Cost (£)
	80	0	0	80

Type of cost: HEI
Description: Subsistence
Conference: False
Item Description: Meals associated with Southampton Developing and Evaluating Complex Interventions course
Justification of Cost: Please see justification for this course in the 'Training and Development' section. This cost is to cover meal allowance (as per King's College London travel policy) for the 4 days associated with this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	100	0	0	100

	Year 1	Year 2	Year 3	Total Cost (£)
Travel, Subsistence and Conference Fees Total	2199	1551	2116	5866

Equipment

Type of cost: HEI
VAT relief to be applied: True

Description: Laptop and monitor
Price excluding VAT £: 983
Expected life time: 60
Justification of Cost: This is to cover the cost of a laptop for the Lead Applicant to allow multisite working (e.g. St Thomas' Hospital and King's College London). This laptop will have all the software programmes on it for data collection and analysis (REDCap, Nutritics, SPSS and ultrasound analysis). A monitor has also been included in the cost to support a multi-screen working setup. The quote was obtained from King's College London IT approved hardware list.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	944	0	0	944

Type of cost: HEI
VAT relief to be applied: True
Description: Hand-held grip dynamometer
Price excluding VAT £: 250
Expected life time: 120
Justification of Cost: This is to cover the cost of purchasing a hand-held grip dynamometer for handgrip strength assessment of participants as part of the research proposal. Please see justification for this assessment method in the 'Detailed Research Plan'. This piece of equipment is not currently available for use in the research laboratory and therefore I will purchase this equipment to allow the study to be conducted as proposed.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	240	0	0	240

Type of cost: HEI
VAT relief to be applied: True
Description: Finger pulse oximeter (SpO2)
Price excluding VAT £: 50
Expected life time: 60
Justification of Cost: This is to cover the cost of two finger pulse oximeters to assess heart rate and oxygen saturation as part of the protocol for the 6-minute step test (Arcuri et al, 2016). This piece of equipment is not available in the research laboratory and therefore I will purchase this equipment to allow the 6-minute step test to be conducted as per the validated protocol.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	48	0	0	48

Type of cost: HEI
VAT relief to be applied: True
Description: BIA machine (Bodystat Quadscan 4000)
Price excluding VAT £: 3250
Expected life time: 60
Justification of Cost: This is to cover the cost of purchasing a bioelectrical impedance analysis (BIA) machine and electrodes for assessment of participants' body composition as part of the research proposal. Please see justification for this assessment method in the 'Detailed Research Plan'. The Bodystat Quadscan 4000 has been selected as it has previously been validated in children and adolescents with PKU (Evans et al, 2017). This piece of equipment is not currently available for use in the research laboratory and therefore I will purchase this equipment to allow the study to be conducted as proposed. The cost is for a starter package which includes: BodyStat Quadscan 4000 device, electrodes (adequate supply for the research proposed), user guide, calibrator checker, and Body Manager Platinum Windows Software.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	3120	0	0	3120

Type of cost: HEI
VAT relief to be applied: True
Description: Skinfold calipers
Price excluding VAT £: 205

Expected life time: 120
Justification of Cost: This is to cover the cost of skinfold calipers for skinfold thickness assessment of participants as part of the research proposal. Please see justification for this assessment method in the 'Detailed Research Plan'. This piece of equipment is not currently available for use in the research laboratory and therefore I will purchase this equipment to allow the study to be conducted as proposed.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	197	0	0	197

Type of cost: HEI
VAT relief to be applied: True
Description: Nutritics analysis package
Price excluding VAT £: 840
Expected life time: 24
Justification of Cost: This is to cover the cost of purchasing a Nutritics software package for food diary analysis as part of the proposed research. The cost will cover purchasing the 'Plus' package for health professionals found on www.nutritics.com/p/pricing. This package has been selected as it allows for appropriate data storage required for the studies and ability to export data to Excel.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	0	403	403	806

Type of cost: HEI
VAT relief to be applied: True
Description: Accelerometers
Price excluding VAT £: 1815
Expected life time: 60
Justification of Cost: This is to cover the cost of purchasing five accelerometer devices for assessment of participants' physical activity as part of the research proposal. Please see justification for this assessment method in the 'Detailed Research Plan'. The activPAL4™ micro has been selected as it accurately quantifies free-living activities including sedentary behaviour. The cost is for a starter package which includes: 5 x activPAL4™ micro devices, PALpatches for attaching the accelerometer device to participants, USB post docking station for charging and data download, and 24 months warranty.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	1742	0	0	1742

Type of cost: HEI
VAT relief to be applied: False
Description: Laboratory and equipment hire
Price excluding VAT £: 1882
Expected life time: 24
Justification of Cost: This is to cover the cost of hiring the research laboratory and equipment (DXA, KinKom) at the Centre for Human and Applied Physiological Sciences (CHAPS), King's College London, that will be used for both the observational and feasibility studies. Please see justification for the use of this research laboratory and equipment in the 'Detailed Research Plan'. The hire costs have been provided by CHAPS Technical Manager.

	Year 1 (exc VAT)	Year 2 (exc VAT)	Year 3 (exc VAT)	Total Cost (£) (exc VAT)
	941	941	0	1882

Type of cost: HEI
VAT relief to be applied: True
Description: Ultrasound equipment (leasing)
Price excluding VAT £: 19080
Expected life time: 36
Justification of Cost: This cost is to cover the leasing of equipment required to perform the ultrasound assessments including: LOGIQ e-portable cart, 9L-RS linear array transducer, 12L-RS linear probe and software. The ultrasound machine selected is one that is appropriate for assessing bicep, forearm and quadriceps muscle layer thickness and rectus femoris cross-sectional area in adult populations with variable muscle mass and body

mass index. Please see justification for this assessment method in the 'Detailed Research Plan'. The quote was obtained from NHS Supply Chain and includes: Purchase cost: £19,729.11 excluding VAT, Indicative residual value lease cost (3 year lease with annual payments): £19,080.10 excluding VAT, Indicative non-residual value lease cost (3 year lease with annual payments): £20,954.09 excluding VAT.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	6106	6106	6106	18318

Type of cost: HEI
VAT relief to be applied: True
Description: Exercise bands for exercise intervention
Price excluding VAT £: 263
Expected life time: 36
Justification of Cost: This is to cover the cost of purchasing exercise bands for participants to use during the feasibility study. Please see justification for this intervention in the 'Detailed Research Plan'. Costing is based on the provision of ~2m of each of the following: thin, medium and heavy Theraband resistance bands per participant.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	252	0	0	252

Type of cost: HEI
VAT relief to be applied: True
Description: Vertical force platform
Price excluding VAT £: 4984
Expected life time: 60
Justification of Cost: This is the cost of purchasing a laboratory force plate, which is considered the gold standard for measuring vertical jump force. This equipment will be used as part of the research proposal. Please see justification for this assessment method in the 'Detailed Research Plan'. This piece of equipment is not currently available for use in the research laboratory and therefore I will purchase this equipment to allow the study to be conducted as proposed. The quote was provided by MedFac UK Ltd for purchase of the K-Deltas Force plate, designed for complex movement and jump analysis. This force plate was selected in consultation with Dr Oliver Witard (Supervisor and Exercise Physiologist) due to its cost effectiveness and high intra- and inter-test reliability.

	Year 1 (inc VAT)	Year 2 (inc VAT)	Year 3 (inc VAT)	Total Cost (£) (inc VAT)
	4785	0	0	4785

	Year 1	Year 2	Year 3	Total Cost (£)
Equipment Total	18375	7450	6509	32334

Consumables

Description: Printing
Type of cost: HEI
Please describe fully: Printing
Justification of Cost: This cost is to cover printing that will be required for the research project. This cost includes printing of consent forms, patient information sheets, questionnaires, calendars, journals, weekly logs and exercise programme for the studies. The quote for printing was obtained from the Reprographics service at St Thomas' Hospital.

	Year 1	Year 2	Year 3	Total Cost (£)
	113	56	0	169

Description: Postage
Type of cost: HEI
Please describe fully: Postage
Justification of Cost: This cost is to cover postage for the following activities: Posting study information and resources (£208): First class stamps and envelopes for sending out study information to eligible participants,

consent forms, pre-paid envelopes to return consent forms and questionnaires. Costing allows for 2 x stamps and envelopes for each participant in the observational study (50 participants) and in the feasibility study (24 participants). Postage of protein substitutes (£607): Costing is based on a quote obtained from DHL website: 7kg parcel (Box 4, dimensions 34 x 32 x 18 cm) costs £16.85. Each participant will require 3 x 7kg boxes of protein substitutes for the feasibility study (total of 12 participants = £606.60). Postage of accelerometers (£165): Cost of posting (tracked delivery and signature on delivery) the accelerometers to study participants in the observational study. Costing is based on pricing from Royal Mail for 'Royal Mail Tracked 24' service (https://www.royalmail.com/sending/uk) = £3.30 per participants

	Year 1	Year 2	Year 3	Total Cost (£)
	677	304	0	981

Description: Blood sample collection and analysis
Type of cost: NHS
Please describe fully: Blood sample collection and analysis
Justification of Cost: This is to cover the costs for blood and bloodspot sample collection, processing and analysis as part of the observational and feasibility studies. The quote was obtained from Guy's and St Thomas' (GSTT) pathology Trust representative and Synnovis, and includes cost for equipment, preparation, storage and testing for 50 blood samples and 100 bloodspots for phenylalanine analysis for the observational study and 24 blood samples and 144 bloodspots for phenylalanine analysis for the feasibility study. Please see the 'Detailed Research Plan' section for justification of biochemical markers. The type of cost is NHS as the funds will be paid directly from King's College London (KCL) to GSTT and Synnovis as part of the grant collaboration agreement between the host (KCL) and partner (GSTT) organisations.

	Year 1	Year 2	Year 3	Total Cost (£)
	4295	4295	0	8590

Description: Protein substitutes
Type of cost: HEI
Please describe fully: Protein substitutes
Justification of Cost: This is the cost to cover purchasing of protein substitutes through procurement at Guys' and St Thomas' Hospital for the feasibility study dietary intervention. Please see justification for this intervention in the 'Detailed Research Plan'. Twelve participants will be in the intervention arm requiring on average additional 30g protein equivalent from protein substitutes per day for 12 weeks. The average cost of 30g protein equivalent from protein substitute based on costings provided from companies and on BNF = £19.45 (£1634 per participant).

	Year 1	Year 2	Year 3	Total Cost (£)
	9802	9802	0	19604

Description: Subsistence for study participants
Type of cost: HEI
Please describe fully: Subsistence for study participants
Justification of Cost: This is to cover the cost of a PKU appropriate breakfast for participants after attending the test days for the observational (50 participants) and feasibility (24 participants) studies. This is costed at £3 per participant (breakfast will consist of 1 x low protein yoghurt, fruit and low protein breakfast bar).

	Year 1	Year 2	Year 3	Total Cost (£)
	150	72	0	222

	Year 1	Year 2	Year 3	Total Cost (£)
Consumables Total	15037	14529	0	29566

Patient and Public Involvement

Description: Phenylketonuria Research Advisory Group involvement
Type of cost: HEI
Please describe fully: Phenylketonuria Research Advisory Group involvement
Justification of Cost: Costing based on NIHR involvement guidance. Payment for PPIE group individuals will be £50 per person per meeting. Meetings will continue to be held via Microsoft teams and run for 1 hours with some pre-meeting prep required. Costing based on payment for 8 members to attend 2 x meetings per year.

	Year 1	Year 2	Year 3	Total Cost (£)

	800	800	800	2400

Description: Project Management Group: Patient and Public Involvement
Type of cost: HEI
Please describe fully: Project Management Group: Patient and Public Involvement
Justification of Cost: Costing is based on NIHR involvement guidance. Payment is for three PPIE members to attend the Project Advisory Group meetings which will occur at the start, mid-point and end of each study (total of six meetings) - £25 per member per meeting, for involvement in a one hour meeting that requires little or no preparation. Meetings will be held via Microsoft teams, therefore no travel is required.

	Year 1	Year 2	Year 3	Total Cost (£)
	225	225	0	450

Description: Compensation for participant involvement in research
Type of cost: HEI
Please describe fully: Compensation for participant involvement in research
Justification of Cost: This cost is to cover compensation for participants (time and travel) to attend the test days for the observational (50 participants) and feasibility (24 participants) studies. Based on NIHR guidance (https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392): £125 per participant (Up to £50 compensation for travel and £75 for participant's time to attend a half day testing session).

	Year 1	Year 2	Year 3	Total Cost (£)
	7750	1500	0	9250

Description: Compensation for participant involvement in the focus group
Type of cost: HEI
Please describe fully: Compensation for participant involvement in the focus group
Justification of Cost: This cost is to cover compensation for service users' participation in the focus group as part of the feasibility study. Based on NIHR guidance (https://www.nihr.ac.uk/documents/payment-guidance-for-researchers-and-professionals/27392): £75 per participant to attend a focus group.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	450	450

	Year 1	Year 2	Year 3	Total Cost (£)
Patient and Public Involvement Total	8775	2525	1250	12550

Other Direct Costs

Description: Poster printing for dissemination
Type of cost: HEI
Please describe fully: Poster printing for dissemination
Justification of Cost: This cost is to cover A0 colour poster printing for presentation at the conferences I will be attending. Costs were obtained from the Reprographics Department at Guy's and St Thomas' Hospital who provide services for poster printing.

	Year 1	Year 2	Year 3	Total Cost (£)
	13	13	13	39

Description: Archiving fee
Type of cost: NHS
Please describe fully: Archiving fee
Justification of Cost: This is to cover the archiving fee for the R&D Department at Guy's and St Thomas' NHS Foundation Trust (GSTT, partner organisation). The quote was obtained from the R&D Department at GSTT. The type of cost is NHS as the funds will be paid directly from King's College London to GSTT as part of the grant collaboration agreement between the host (KCL) and partner (GSTT) organisations.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	0	565	565

Description: Electronic database (REDCap)
Type of cost: HEI
Please describe fully: Electronic database (REDCap)
Justification of Cost: This is to cover the cost of the online electronic data capturing system, REDCap, which will be used to ensure data is accurately captured and to allow for more efficient data analysis. REDCap also includes online survey features which will be used for the feasibility study. REDCap is a secure web application for building and managing online surveys and databases, which is approved by KCL and GSTT, being currently used across both sites. There are self-training videos available on setting up, building and managing the database. In addition, support for trouble shooting is available through KCL/GSTT. There are further plans to link REDCap to EPIC, the electronic patient management system at GSTT, allowing for patient data to be directly transferred from EPIC to REDCap. There is an annual cost of £2,500 per year to have access to REDCap.

	Year 1	Year 2	Year 3	Total Cost (£)
	2500	2500	2500	7500

Description: HRA Radiation Assurance fee
Type of cost: HEI
Please describe fully: HRA Radiation Assurance fee
Justification of Cost: HRA Radiation Assurance will soon become the expected route for all studies involving ionising radiation. Any research studies involving ionising radiation that are being undertaken or partnered with GSTT are required to cost for HRA Radiation Assurance. This has also been requested by King’s College London Centre of Human and Applied Physiological Science (CHAPS) where my research will be conducted. As the research proposal includes DXA scans, I have costed for HRA Radiation Assurance. Radiation Assurance takes place before submission of the IRAS form to ethics/HRA.

	Year 1	Year 2	Year 3	Total Cost (£)
	1000	0	0	1000

Description: CTU Support Services: Randomisation
Type of cost: HEI
Please describe fully: CTU Support Services: Randomisation
Justification of Cost: This cost is to cover randomisation services provided by the King's Clinical Trials Unit (KCTU). Randomisation will be required for the feasibility study, see 'Detailed Research Plan'. The quote has been provided by KCTU, see 'Uploads'.

	Year 1	Year 2	Year 3	Total Cost (£)
	650	0	0	650

Description: Licence fee for force platform
Type of cost: HEI
Please describe fully: Licence fee for force platform
Justification of Cost: This is the cost to cover the licence fee for the K-Deltas Force plate app software. The force plate will be used to assess vertical jump force as part of the research proposal. Annual licence fee is £245.75, and the study will require the licence for 2 years.

	Year 1	Year 2	Year 3	Total Cost (£)
	246	246	0	492

	Year 1	Year 2	Year 3	Total Cost (£)
Other Direct Costs Total	4409	2759	3078	10246

Training And Development

Type of cost: HEI
Description: Applicant PhD Tuition Fees
Item Description: Applicant PhD Tuition Fees
Justification of Cost: This cost is to cover my PhD Tuition Fees at King's College London. The costs are the projected tuition fees based on the 2023/2024 full time home fees for the academic year.

	Year 1	Year 2	Year 3	Total Cost (£)

	7919	7919	7919	23757

Type of cost: HEI
Description: Short Courses
Item Description: Oxford University RCT course
Justification of Cost: Course fee (based on 2024 fees) for the Oxford Randomised Controlled Trials Course: A guide to design, conduct, analysis, interpretation and reporting. Please see the 'Training and Development' section for the justification of this course. Cost includes daily lunch, a course dinner, and a comprehensive delegate's pack including all course materials.

	Year 1	Year 2	Year 3	Total Cost (£)
	1550	0	0	1550

Type of cost: HEI
Description: Short Courses
Item Description: ISAK Level 1 anthropometry course
Justification of Cost: Course fees to attend Level 1 ISAK Anthropometry Course offered by St Mary's University. Please see the 'Training and Development' section for the justification of this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	420	0	0	420

Type of cost: HEI
Description: Short Courses
Item Description: IR(ME)R certification and training
Justification of Cost: Vertec Scientific Limited provides training to all researchers conducting DXA scans in the Centre for Human and Applied Physiological Sciences (CHAPS), King's College London. It is a requirement to have this training prior to using the scanner at CHAPS. The training includes practical training on using the scanner including calibration. I will be conducting DXA scans as part of my research proposal. Please see justification for this assessment method in the 'Detailed Research Plan'. The training course allows attendees to develop competence and demonstrate compliance with the regulatory requirements of the IR(ME)R.

	Year 1	Year 2	Year 3	Total Cost (£)
	1210	0	0	1210

Type of cost: HEI
Description: Short Courses
Item Description: Practical Statistics for Medical Research
Justification of Cost: Course fees to attend 'Practical Statistics for Medical Research' offered by University College London. Please see the 'Training and Development' section for the justification of this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	550	0	0	550

Type of cost: HEI
Description: Short Courses
Item Description: Developing and Evaluating Complex Interventions: The University of Southampton
Justification of Cost: Course fees to attend 'Developing and Evaluating Complex Interventions' offered by University of Southampton. Please see the 'Training and Development' section for the justification of this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	225	0	0	225

Type of cost: HEI
Description: Short Courses
Item Description: Questionnaire design
Justification of Cost: Course fees to attend the online Questionnaire design (NatCen: Social Research Association) course. Please see the 'Training and Development' section for the justification of this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	220	0	0	220

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to Qualitative Interviewing
Justification of Cost: Course fees to attend the online Introduction to Qualitative Interviewing course offered by Social Research Association. Please see the 'Training and Development' section for the justification of this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	220	0	0	220

Type of cost: HEI
Description: Short Courses
Item Description: Introduction to focus groups
Justification of Cost: Course fees to attend the online Introduction to focus groups course offered by Social Research Association. Please see the 'Training and Development' section for the justification of this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	220	0	220

Type of cost: HEI
Description: Short Courses
Item Description: Qualitative Data Analysis
Justification of Cost: Course fees to attend the online Qualitative Data Analysis course offered by Social Research Association. Please see the 'Training and Development' section for the justification of this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	0	220	0	220

Type of cost: HEI
Description: Short Courses
Item Description: Phlebotomy course
Justification of Cost: Course fees to attend a London based phlebotomy training course hosted by Phlebotomy Training Services. Please see the 'Training and Development' section for the justification of this course.

	Year 1	Year 2	Year 3	Total Cost (£)
	348	0	0	348

	Year 1	Year 2	Year 3	Total Cost (£)
Training And Development Total	12662	8359	7919	28940

Total Direct Costs

	Year 1	Year 2	Year 3	Total Cost (£)
Total Direct Costs	137351	114879	93629	345859

Indirect costs

HEI Indirect costs

Type of cost: Estates Charges
Name of Higher Education Institute: King's College London
Justification of Cost: The following costs have been provided by King's College London Research Finance using TRAC/FEC methodology. Estates charges also include infrastructure technicians.

	Year 1	Year 2	Year 3	Total Cost (£)
	26751	26751	26751	80253

Type of cost: Other Indirect Costs
Name of Higher Education Institute: King's College London
Justification of Cost: The following costs have been provided by King's College London Research Finance using TRAC/FEC methodology.

	Year 1	Year 2	Year 3	Total Cost (£)
	69745	69745	69745	209235

	Year 1	Year 2	Year 3	Total Cost (£)
HEI Indirect costs Total	96496	96496	96496	289488

Commercial Indirect Costs
No costs have been added

Other Partner Organisation Indirect costs
No costs have been added

	Year 1	Year 2	Year 3	Total Cost (£)
Total Indirect Costs	77197	77197	77197	231591

Research Costs Requested From Funder

	Direct costs	Indirect costs/Overheads	Amount requested
Total Higher Education Institution Costs (80%)	68817	231591	300408
Total Higher Education Institution Costs (100%)s	31940	0	31940
Total Commercial Costs	0	0	0
Total Other Parther Organisation Costs	0	0	0
Total NHS Research Costs	245102	0	245102
Total Research Costs Requested From Funder	345859	231591	577450

Summary

	Year 1	Year 2	Year 3	Total Cost (£)
Salaries	**81107**	**81107**	**73733**	**235947**
Travel-subsistence-and-conference-fees	**1959**	**1441**	**1893**	**5293**
Equipment	**18187**	**7262**	**6509**	**31958**
Consumables	**12889**	**12483**	**0**	**25372**
Patient-and-Public-Involvement	**7020**	**2020**	**1000**	**10040**
Other-Direct-Costs	**3527**	**2207**	**2575**	**8309**
Training And Development	**12662**	**8359**	**7919**	**28940**
HEI-Indirect-costs	**77197**	**77197**	**77197**	**231591**

Commercial-Indirect-Costs	**0**	**0**	**0**	**0**
Other-Partner-Organisation-Indirect-costs	**0**	**0**	**0**	**0**
Total	**214548**	**192076**	**170826**	**577450**

NHS Support Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
NHS Service Support Costs	4818	0	0	4818.0

Justification:
NHS Service Support Costs: This is the cost to cover the time spent reviewing patients' medical records for screening eligible participants, recruiting and consenting to the studies (as outlined in the SoECAT).

	Year 1	Year 2	Year 3	Total Cost (£)
NHS Support Costs Total	4818	0	0	4818

Have you discussed and agreed these costs with the Lead Network?	Yes

NHS Treatment Costs

No costs have been added

Have you discussed and agreed the NHS costs with the Lead Trust?	N/A

Is the patient/service user care being provided different from the usual standard treatment for the condition?
No

Usual Treatment Costs

Cost Item:	Year 1	Year 2	Year 3	Total Cost (£)
Usual Treatment Costs	0	0	0	0.0

Justification:
Usual Treatment Costs: The study intervention is in addition to usual routine patient care and therefore no 'Usual Treatment Costs' have been included. Patients will continue to receive their usual patient care as provided by the Adult Inherited Metabolic Diseases Services at Guy's and St Thomas' Hospital.

	Year 1	Year 2	Year 3	Total Cost (£)
Usual Treatment Costs Total	0	0	0	0

Excess Treatment costs/(savings)

	Cost per patient(£)	Year 1	Year 2	Year 3	Total Cost (£)
Total NHS Excess Treatment costs/(savings)	0	0	0	0	0

	Total (£)
NHS Support Costs requested from networks	4818
NHS Excess Treatment costs/savings	0
TOTAL NHS SUPPORT AND TREATMENT COSTS	4818

Total Funding Required (including non-research costs)

	Total (£)
Total Research Costs Requested from Funder	577450
Total NHS (Support and Excess Treatment) Costs	4818
Total Funding Required	582268

10. Uploads

Please upload the following documents, if necessary.

When uploading a document to your application, please ensure that it is in an appropriate file format. It is not possible to upload JPEG images, and with the exception of the SoECAT Form, which should be uploaded as an Excel spreadsheet, all files should be uploaded in Adobe PDF or Word DOC/DOCX format. Please refer to the Guidance Notes for further information on the attachments required.

(References - SFirman_References.pdf) is included as an appendix within this file.

(Figures / Tables - SFirman_Figures&Table.pdf) is included as an appendix within this file.

(Research Timetable - SFirman_Research and training timetable.pdf) is included as an appendix within this file.

(CTU Letter of Support - CTU Letter of Support.pdf) is included as an appendix within this file.

(SoECAT Template - Funder Export_344077_SoECAT_Muscle mass and muscle function in Phenylketonuria.xlsx) is included as an appendix within this file.

11. Participants and Signatories

Participants
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required participant. Everyone included in this section will be acting as a 'participant' in your application and must agree to be named as such.

By confirming participation, participants are acknowledging their involvement and input into this application, and are agreeing to be involved, as indicated, in any resultant fellowship.

Once the 'Administrative Authority or Finance Officer' participant has confirmed their participation, they will be able to access the 'Detailed Budget' section of the application form. You should collaborate with your 'Administrative Authority or Finance Officer participant to ensure that this section is completed satisfactorily.

You must ensure that all participants are happy for your application to be submitted before you submit it.

It is permissible for a single individual to serve as both a signatory and a participant, but they will need to be added separately under each heading.

Named participants must confirm their participation on your application before you will be able to select the 'SUBMIT' option. Please refer to the Submission Process Flow Diagram and the Applicant Guidance Notes for further information.

Primary Doctoral Supervisor

Doctoral Primary Supervisor Participant 1

Title	Professor
Forename(s)	Kevin
Surname	Whelan
Position	Professor of Dietetics

Additional Academic Supervisor(s)

Supervisor Participant 1

Title	Dr
Forename(s)	Radha
Surname	Ramachandran
Position	Clinical Lead and Consultant in Adult Inherited Metabolic Diseases

Supervisor Participant 2

Title	Dr
Forename(s)	Oliver
Surname	Witard

Position	Senior Lecturer

Practice Supervisor(s)

Practice Support Participant 1

Title	Dr
Forename(s)	Danielle
Surname	Bear
Position	Consultant Critical Care Dietitian

Practice Support Participant 2

Title	Dr
Forename(s)	Radha
Surname	Ramachandran
Position	Clinical Lead and Consultant in Adult Inherited Metabolic Diseases

Administrative Authority or Finance Officer

Administrative Authority or FO Participant

Title	Ms
Forename(s)	Elizabeth
Surname	Fell
Position	

Signatories
You are required to supply the name and email address (if not already registered on the ARAMIS application system) of each required ‘Head of Department / Senior Manager’ signatory. Once their contact details have been entered below, these signatories can be invited to log in and confirm their participation in your application.

Once the nominated signatories have confirmed their participation, they can access and complete the required ‘Host Organisation Statement of Support’ in the ‘Training & Development and Research Support’ section of the application form.

Your signatories must confirm their participation in your application and complete the section they are responsible for before you can select the ‘submit’ option.

Signatories must also approve your application after you select the 'submit' option but BEFORE the application submission deadline.

Signatory approval from your 'Head of Department / Senior Manager' signatories will result in the application being fully submitted to the NIHR. All parties (applicant, participants and signatories) will be notified of this via an automated system-generated email.

The NIHR will not accept your application unless fully approved by your signatories before the 1 p.m. deadline on Tuesday 11th June 2024.

Heads of Departments / Senior Managers

Head of Department or Senior Manager 1

Title	Professor
Forename(s)	Kevin
Surname	Whelan
Position	Professor of Dietetics

Head of Department or Senior Manager 2

Title	Mr
Forename(s)	Rhys
Surname	White
Position	

12. Acknowledgement, Review and Submit

Conflict checks: Please declare any conflicts or potential conflicts of interest that you may have in undertaking this research, including any relevant, non-personal & commercial interest that could be perceived as a conflict of interest.
I have received funding to attend conferences and study days from Nutricia, Vitaflo International and Dr. Schär UK Ltd, and consulting fees from Vitaflo International and Meta Healthcare Ltd. These companies produce protein substitutes and specialist low protein food products for PKU.

By completing this form, the programme reserves the right to share, in confidence, details of your application with other parts of the NIHR and with the Department of Health and Social Care.

Agreement to Terms and Conditions:

I confirm that I have secured all necessary licences and ethical approvals in relation to the research and will abide by the terms of those licences and approvals in carrying out the research.

I confirm that I have read the NIHR Carbon Reduction Guidelines and, where possible, have taken steps to reduce the carbon emissions generated by this research.

I confirm that all material within this application, with the exception of sections that other participants are required to complete, is my own work and I have taken steps to ensure that no parts of the application have been inappropriately plagiarised.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time lay down according to the NIHR research governance framework.

I undertake to conduct the research according to such guidelines for good research practice as the Host Organisation may from time to time laydown according to the UK Policy Framework for Health and Social Care Research.

In completing this, I declare that, to the best of my knowledge, the information provided in this application, is true, accurate and complete. I have read the Guidance Notes and agree to accept the process by which an application is assessed and agree to abide by the conditions under which an award may be granted.

☒Confirmed

Appendices

1. SFirman_References.pdf
2. SFirman_Figures&Table.pdf
3. SFirman_Research and training timetable.pdf
4. CTU Letter of Support.pdf
5. Funder Export_344077_SoECAT_Muscle mass and muscle function in Phenylketonuria.xlsx

Appendix #1: SFirman_References.pdf

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Appendix #2: SFirman_Figures&Table.pdf

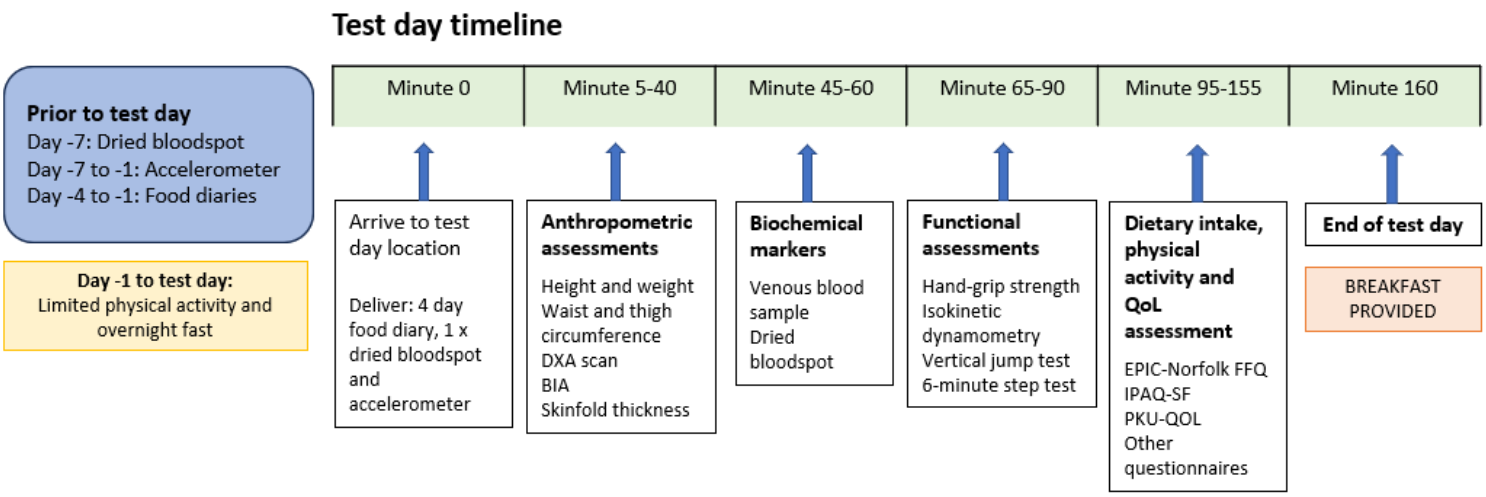

Figure 1: Observational study design and outcome measurements

DXA: dual energy X-ray absorptiometry, BIA: bioelectrical impedance analysis, EPIC-Norfolk FFQ: European Prospective Investigation of Cancer in Norfolk Food Frequency Questionnaire, IPAQ-SF: International Physical Activity Questionnaire - Short Form, PKU-QOL: Adult Health-related Quality of Life questionnaire

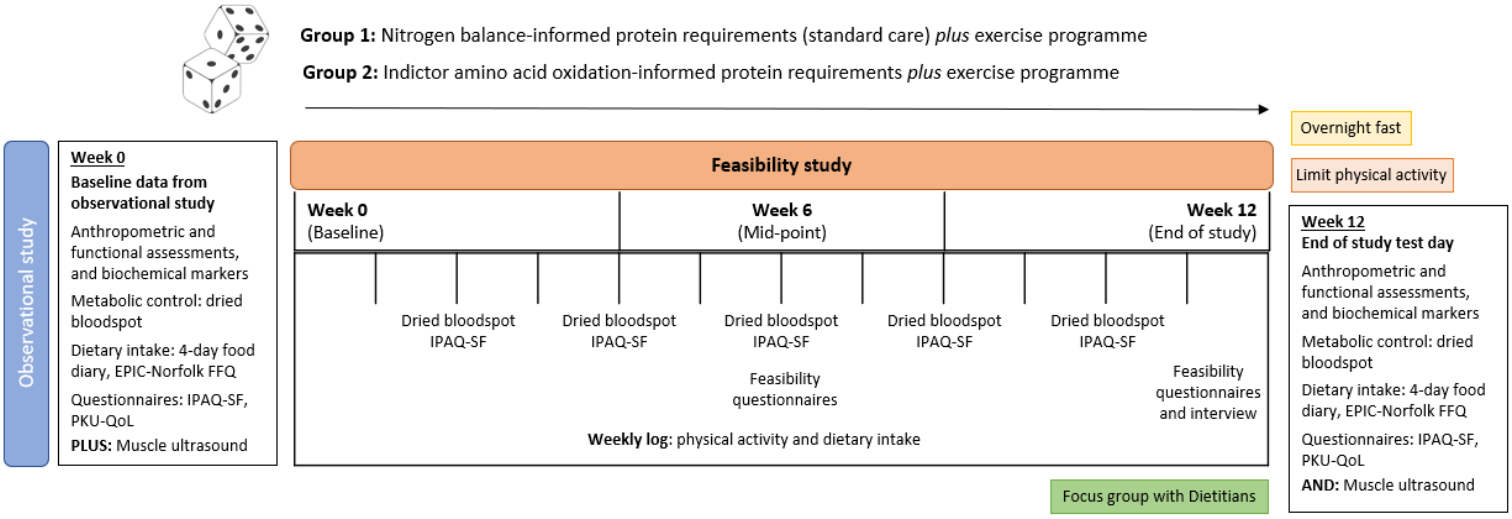

Figure 2: Feasibility study design and outcome measurements

EPIC-Norfolk FFQ: European Prospective Investigation of Cancer in Norfolk Food Frequency Questionnaire, IPAQ-SF: International Physical Activity Questionnaire - Short Form, PKU-QOL: Adult Health-related Quality of Life questionnaire

Table 1: Progression criteria to full trial

Criteria	Green	Amber	Red
Feasibility of recruiting to target	100%	60-99%	<60%
Retention - Attendance at week 12 test day	100%	60-99%	<60%
Intervention fidelity:			
Adherence to dietary intervention	>75%	40-75%	<40%
% of participants completing IPAQ-SF	>75%	40-75%	<40%
Adherence to exercise intervention	>75%	40-75%	<40%
Acceptability of Intervention Measure	Mean score 20	Mean score 10-19	Mean score <10
Feasibility Intervention Measure	Mean score 20	Mean score 10-19	Mean score <10

Green: Progress to full-scale trial with screening log review and protocol revision to address any barriers to recruitment, retention and intervention deliverability.

Amber: Consider progression to full-scale trial with further consultation with key stakeholders and PPIE, and review of screening logs and protocol addressing recruitment, retention and intervention deliverability barriers and harnessing facilitators.

Red: Unable to progress to full-scale trial

Appendix #3: SFirman_Research and training timetable.pdf

Research and training timetable (Gantt chart)

Year	2025								2026												2027												2028							
Month	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A	S	O	N	D	J	F	M	A	M	J	J	A
Month number	-4	-3	-2	-1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36
Research																																								
Ethics / R&D approval																																								
PPIE meetings																																								
Project Advisory Group meetings																																								
Purchasing study equipment/stock																																								
Study 1: Recruitment and data collection																																								
Study 2: Recruitment, trial & data collection																																								
Study 1: Data analysis																																								
Conduct focus group																																								
Study 2: Transcribing and data analysis																																								
Write-up thesis and dissemination																																								
Training & development																																								
Research governance training																																								
RCT course (Oxford)																																								
ISAK training																																								
IR(ME)R training																																								
Questionnaire design training																																								
PPIE training																																								
Phlebotomy course																																								
Complex intervention course																																								
Conferences																																								
NIHR leadership modules																																								
Qualitative interviewing training																																								
Statistics course																																								
SPSS/R training																																								
Qual analysis course																																								
Focus group training																																								
Site visit (Birmingham)																																								
Overseas visit																																								
Professional training, and development																																								
KCL doctoral training courses																																								

Appendix #4: CTU Letter of Support.pdf

King's Clinical Trials Unit (KCTU)

NIHR | Research Support Service

This unit receives National Institute for Health Research RSS Funding.

Professor Richard Emsley, KCTU Director
Caroline Murphy, KCTU Operational Director

King's Clinical Trials Unit • PO Box 64 • 16 De Crespigny Park • Denmark Hill • London SE5 8AF
TEL: + 44 (0)20 7848 0532 • FAX: +44 (0)20 7848 5229 • www.ctu.co.uk

DATE 17-05-24

Dear Dr. Firman,

This is to confirm that the King's Clinical Trials Unit, King's College London (UKCRC registration number 053) have considered your request for support for;

Study title: Impairment of muscle mass, muscle function and quality of life in phenylketonuria: optimising protein intakes safely, effectively and feasibly.
Funding Stream: Doctoral Fellowship
Application Stage: One stage

The CTU will provide:

Service/Collaboration	Service cost or collaborator name	Type of service or FTE for collaborator
Randomisation service	£650 excl. VAT	Service type Bronze

We will be delighted to support you in this bid.

Please note prices above are not inclusive of VAT.

Yours sincerely

Caroline Murphy

Operational Director, King's Clinical Trials Unit

Appendix #5: Funder Export_344077_SoECAT_Muscle mass and muscle function in Phenylketonuria.xlsx